

# DESIGN OF EXPERIMENTS

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# Design of experiments

- Usually modeling (i.e. simulation) is carry out as a programming exercise.
- Inaccurate statistical methods (no IID).
- Take care of the time required to collect the needed data to apply the statistical techniques, with guaranties of achieve the accomplishment of the objectives.

# Design of experiments

- How to make the **comparisons** between different configurations.
  - ▣ The comparisons must be the more homogeneous as possible.
- Study the **effect** over the answer variable of the values of the different experimental variables.
  - ▣ In a cashier: Answer variable: Queue long; factors: Number of cashiers, service time, time between arrivals.

# Principles

Principles to develop a good design of experiments:

- **Randomization:** Assignment to the random of all the factors that are not controlled by the experimentation.
- **Repetition of the experiment (replication):** Is a good method to reduce the variability between the answers.
- **Statistical homogeneity of the answers:** To compare different alternatives derived from the results, is needed that the executions of the experiments have been done under homogeny conditions. Factorial design helps to obtain this similarity between the experiments.

# Design of Experiments goals

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- **Isolate** effects of each input variable.
- Determine **effects** of interactions.
- Determine **magnitude** of experimental error
- Obtain maximum **information** for given effort

# Terminology

- Response variable

  - ▣ Measured output value

    - E.g. total execution time

- Factors

  - ▣ Input variables that can be changed

    - E.g. cache size, clock rate, bytes transmitted

- Levels

  - ▣ Specific values of factors (inputs)

    - Continuous (~bytes) or discrete (type of system)

# Terminology

- Replication
  - ▣ Completely re-run experiment with same input levels
  - ▣ Used to determine impact of measurement error
- Interaction
  - ▣ Effect of one input factor depends on level of another input factor
- Treatment, scenario, one of the combinations to analyze.

# Tests

| Test                                           | Dependent variables | Independent variables |
|------------------------------------------------|---------------------|-----------------------|
| T-test                                         | One                 | One (two groups)      |
| ANOVA<br>Completely Randomized Design (CRD)    | One                 | One                   |
| Two-way ANOVA<br>Randomized Block Design (RBD) | One                 | Two                   |
| MANOVA                                         | Mutiple             | Multiple              |



| Test # | Test name                                                                                                      | Check      | Statistic | Assumptions | Ind              | $\sigma$ | $\sigma_1 = \sigma_2$      | d | $\mu/p$ | $\bar{x}/\bar{p};$ | n | S |   |
|--------|----------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------------|----------|----------------------------|---|---------|--------------------|---|---|---|
| 1      | <a href="#">One Sample Z-Test</a>                                                                              | Mean       |           | ✓           | ✓                | ✓        |                            |   | ✓       | ✓                  | ✓ |   |   |
| 2      | <a href="#">One Sample T-Test</a>                                                                              | Mean       |           | ✓           | ✓                | ✗        |                            |   | ✓       | ✓                  | ✓ | ✓ |   |
| 3      | <a href="#">Two Sample Z-Test</a>                                                                              | Mean       |           | ✓           | ✓                | ✓        |                            | ✓ |         | ✓                  | ✓ |   |   |
| 4      | <a href="#">Two Sample T-Test (Pooled variance)</a>                                                            | Mean       |           | ✓           | ✓                | ✗        | ✓                          | ✓ |         | ✓                  | ✓ | ✓ |   |
| 5      | <a href="#">Two Sample T-Test (Welch's)</a>                                                                    | Mean       |           | ✓           | ✓                | ✗        | ✗                          | ✓ |         | ✓                  | ✓ | ✓ |   |
| 6      | <a href="#">Two Sample Mann-Whitney U Test</a>                                                                 | Rank       |           |             | ✓                |          |                            | ✓ |         |                    |   |   | ✓ |
| 7      | <a href="#">Paired T-Test</a>                                                                                  | Mean       |           | ✓           | Paired           |          |                            |   |         |                    |   |   | ✓ |
| 8      | <a href="#">Paired Wilcoxon Sign Rank Test</a>                                                                 | Rank       |           |             | Paired           |          |                            |   |         |                    |   |   | ✓ |
| 9      | <a href="#">One Way ANOVA Test</a>                                                                             | Mean       |           | ✓           | ✓                |          | ✓                          |   |         |                    |   |   | ✓ |
| 10     | <a href="#">Repeated Measures ANOVA Test</a>                                                                   | Mean       |           | ✓           | Dependent groups |          | <a href="#">Sphericity</a> |   |         |                    |   |   | ✓ |
| 11     | <a href="#">Kruskal-Wallis Test</a>                                                                            | Mean       |           |             | ✓                |          |                            |   |         |                    |   |   | ✓ |
| 12     | <a href="#">Two Way ANOVA Test</a><br>Fixed model,<br>Mixed model,<br>Random mode,<br>Mixed model with repeats | Mean       |           | ✓           | ✓                |          | ✓                          |   |         |                    |   |   | ✓ |
| 13     | <a href="#">One Way MANOVA Test</a>                                                                            | Mean       | F         | ✓           | ✓                |          | ✓                          |   |         |                    |   |   | ✓ |
| 14     | <a href="#">One Sample Proportion Test</a>                                                                     | Proportion |           | Binomial    | ✓                |          |                            |   | ✓       | ✓                  | ✓ |   |   |
| 15     | <a href="#">Two Sample Proportion Test</a>                                                                     | Proportion |           | Binomial    | ✓                |          |                            |   |         | ✓                  | ✓ |   |   |
| 16     | <a href="#">Chi-Squared Test For Variance</a>                                                                  | $\sigma$   |           | ✓           | ✓                |          |                            |   |         |                    | ✓ | ✓ |   |
| 17     | <a href="#">F Test For Variances</a>                                                                           | $\sigma$   |           | ✓           | ✓                |          |                            |   |         |                    | ✓ | ✓ |   |
| 18     | <a href="#">Levene's Test For Variances</a>                                                                    | $\sigma$   |           | ✗           | ✓                |          |                            |   |         |                    |   |   | ✓ |
| 19     | <a href="#">Chi-Squared Test For Goodness Of Fit</a><br>(McNemar test,<br>Chi-squared test for                 | Fit        |           |             | ✓                |          |                            |   |         |                    |   |   | ✓ |

<https://www.statskingdom.com/index.html>

# One-Factor ANOVA

- Separates total variation observed in a set of measurements into:
  - ▣ Variation within one system
    - Due to random measurement errors
  - ▣ Variation between systems
    - Due to real differences + random error
  
- *One-factor experimental design*
  - ▣ *We have here three or more different populations?*



# Two-factor Experiments

Introduction to Design of Experiments

# Two-factor Experiments

- Two factors (inputs)
  - ▣ A, B
- Separate total variation in output values into:
  - ▣ Effect due to A
  - ▣ Effect due to B
  - ▣ Effect due to interaction of A and B (AB)
  - ▣ Experimental error

# Example – User Response Time

- A = degree of multiprogramming
- B = memory size
- AB = interaction of memory size and degree of multiprogramming

|   | B (Mbytes) |       |       |
|---|------------|-------|-------|
| A | 32         | 64    | 128   |
| 1 | 1,125      | 1,105 | 1,075 |
| 2 | 1,26       | 1,225 | 1,18  |
| 3 | 1,405      | 1,33  | 1,25  |
| 4 | 1,75       | 1,725 | 1,35  |

Factor A

Factor A

Factor A

Factor B

Factor C

$y_{ijk}$

n replications

# One-factor ANOVA

- Each individual measurement is composition of
  - ▣ Overall mean
  - ▣ Effect of alternatives
  - ▣ Measurement errors

$$y_{ij} = \bar{y}_{..} + \alpha_i + e_{ij}$$

$$\bar{y}_{..} = \text{overall mean}$$

$$\alpha_i = \text{effect due to A}$$

$$e_{ij} = \text{measurement error}$$

# Two-factor ANOVA

- Factor A –  $a$  input levels
- Factor B –  $b$  input levels
- $n$  measurements for each input combination
- $abn$  total measurements
- Each individual measurement is composition of
  - ▣ Overall mean
  - ▣ Effects
  - ▣ Measurement errors
  - ▣ Interactions

$$y_{ijk} = \bar{y}_{...} + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}$$

$\bar{y}_{...}$  = overall mean

$\alpha_i$  = effect due to A

$\beta_j$  = effect due to B

$\gamma_{ij}$  = effect due to interaction of A and B

$e_{ijk}$  = measurement error



# Sum-of-Squares

- As before, use sum-of-squares identity

$$SST = SSA + SSB + SSAB + SSE$$

- Degrees of freedom

- $df(SSA) = a - 1$

- $df(SSB) = b - 1$

- $df(SSAB) = (a - 1)(b - 1)$

- $df(SSE) = ab(n - 1)$

- $df(SST) = abn - 1$

$$\underbrace{\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r (Y_{ijk} - \bar{Y}_{...})^2}_{SS_{Total}} = \underbrace{r \cdot b \cdot \sum_{i=1}^a (\bar{Y}_{i..} - \bar{Y}_{...})^2}_{SS_A} + \underbrace{r \cdot a \cdot \sum_{j=1}^b (\bar{Y}_{.j.} - \bar{Y}_{...})^2}_{SS_B} + \underbrace{r \times \sum_{i=1}^a \sum_{j=1}^b (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2}_{SS_{A \times B}} + \underbrace{\sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^r (Y_{ijk} - \bar{Y}_{ij.})^2}_{SS_{within}}$$

$$MS_{within} = SS_{within} / df_{within}$$

# Two-Factor ANOVA table

| Source       | Degrees of Freedom | SS                | MS                | F                               |
|--------------|--------------------|-------------------|-------------------|---------------------------------|
| A            | a-1                | $SS_A$            | $MS_A$            | $MS_A / MS_{within}$            |
| B            | b-1                | $SS_B$            | $MS_B$            | $MS_B / MS_{within}$            |
| $A \times B$ | (a-1)(b-1)         | $SS_{A \times B}$ | $MS_{A \times B}$ | $MS_{A \times B} / MS_{within}$ |
| Within       | ab(r-1)            | $SS_{within}$     | $MS_{within}$     |                                 |
| Total        | abr-1              | $SS_{Total}$      |                   |                                 |

# $n2^k$ Sum of Squares - Contrasts

## □ Effects of A, B and interactions.

- (a) :  $w_A = y_{AB} + y_{Ab} - y_{aB} - y_{ab}$

- (b):  $w_B = y_{AB} - y_{Ab} + y_{aB} - y_{ab}$

- (ab):  $w_{AB} = y_{AB} - y_{Ab} - y_{aB} + y_{ab}$

- $SSA = \frac{w_A^2}{n2^k}$

- $SSB = \frac{w_B^2}{n2^k}$

- $SSAB = \frac{w_{AB}^2}{n2^k}$

- $SSE = SST - SSA - SSB - SSAB$

# Replications are needed

- If no replications,  $n=1$  (only one measurement of each configuration)
  - ▣  $SSE = 0$ , no information regarding the errors in the measurements.
- Since
  - ▣  $SSE = SST - SSA - SSB - SSAB$
- and
  - ▣  $SSAB = SST - SSA - SSB$
- Cannot separate effect due to interactions from measurement noise
- Must *replicate* each experiment at least twice

# Example

- Output = user response time (seconds)
- Want to separate effects due to
  - ▣ A = degree of multiprogramming
  - ▣ B = memory size
  - ▣ AB = interaction
  - ▣ Error
- Need **replications** to separate error

|   | B (Mbytes) |       |       |
|---|------------|-------|-------|
| A | 32         | 64    | 128   |
| 1 | 1,125      | 1,105 | 1,075 |
| 2 | 1,26       | 1,225 | 1,18  |
| 3 | 1,405      | 1,33  | 1,25  |
| 4 | 1,75       | 1,725 | 1,35  |

# Example

| A | B (Mbytes) |       |       |
|---|------------|-------|-------|
|   | 32         | 64    | 128   |
| 1 | 1,125      | 1,105 | 1,075 |
|   | 1,14       | 1,095 | 1,055 |
| 2 | 1,26       | 1,225 | 1,18  |
|   | 1,24       | 1,245 | 1,15  |
| 3 | 1,405      | 1,33  | 1,25  |
|   | 1,38       | 1,295 | 1,305 |
| 4 | 1,75       | 1,725 | 1,35  |
|   | 1,125      | 1,105 | 1,075 |

# Two-Factor ANOVA table example

|                | A                           | B                           | AB                          | Error                    |
|----------------|-----------------------------|-----------------------------|-----------------------------|--------------------------|
| Sum of squares | $SSA$                       | $SSB$                       | $SSAB$                      | $SSE$                    |
| Deg freedom    | 1                           | 1                           | 1                           | $2^m(n-1)$               |
| Mean square    | $s_a^2 = SSA/1$             | $s_b^2 = SSB/1$             | $s_{ab}^2 = SSAB/1$         | $s_e^2 = SSE/[2^m(n-1)]$ |
| Computed $F$   | $F_a = s_a^2/s_e^2$         | $F_b = s_b^2/s_e^2$         | $F_{ab} = s_{ab}^2/s_e^2$   |                          |
| Tabulated $F$  | $F_{[1-\alpha;1,2^m(n-1)]}$ | $F_{[1-\alpha;1,2^m(n-1)]}$ | $F_{[1-\alpha;1,2^m(n-1)]}$ |                          |

| ANOVA               |          |    |          |          |          |          |
|---------------------|----------|----|----------|----------|----------|----------|
| Source of Variation | SS       | df | MS       | F        | P-value  | F crit   |
| Sample              | 0,842854 | 3  | 0,280951 | 460,2617 | 1,2E-12  | 3,490295 |
| Columns             | 0,128802 | 2  | 0,064401 | 105,5034 | 2,43E-08 | 3,885294 |
| Interaction         | 0,107915 | 6  | 0,017986 | 29,46473 | 1,65E-06 | 2,99612  |
| Within              | 0,007325 | 12 | 0,00061  |          |          |          |
|                     |          |    |          |          |          |          |
| Total               | 1,086896 | 23 |          |          |          |          |

# $n2^k$ Contrasts

- Effects of A, B and interactions.
- $W_A = y_{AB} + y_{Ab} - y_{aB} - y_{ab}$
- $W_B = y_{AB} - y_{Ab} + y_{aB} - y_{ab}$
- $W_{AB} = y_{AB} - y_{Ab} - y_{aB} + y_{ab}$

| (A) | (B) | Y        |
|-----|-----|----------|
| A   | B   | $y_{AB}$ |
| A   | b   | $y_{Ab}$ |
| a   | B   | $y_{aB}$ |
| a   | b   | $y_{ab}$ |





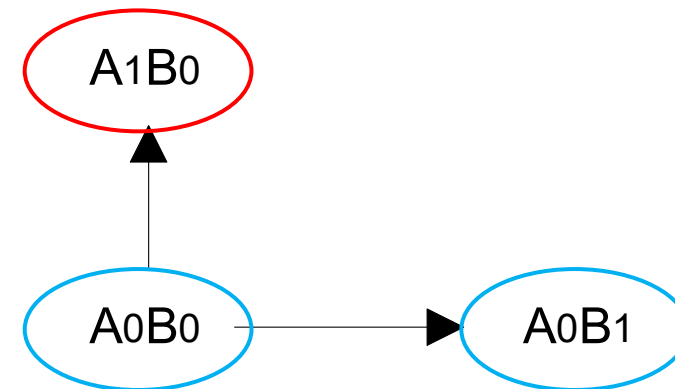
# Factorial designs

Explore the landscape

Assure that we are analyzing all the combinations with economy  
on the experimentation

# No factorial designs

- To fix two factors and modify all the levels of a third until find a good solution. Fixing this level, start the exploration for the other factors.
- Effect A:  $A_1B_0 - A_0B_0$ .
- Effect B:  $A_0B_1 - A_0B_0$



# Factorial designs

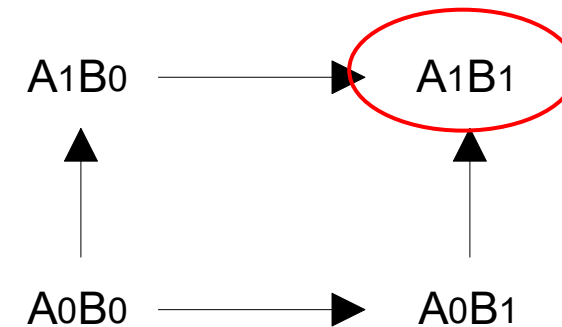
- Take in consideration the interactions.

- A effect:

$$\frac{A_1B_0 - A_1B_1}{2} - \frac{A_0B_0 - A_0B_1}{2}$$

- B effect:

$$\frac{A_1B_1 - A_0B_1}{2} - \frac{A_0B_0 - A_1B_0}{2}$$



# Factorial designs

- Controlling “k” factors.
- “l” levels for each factor (“li” levels for the i factor).
- $l_1 \cdot l_2 \cdot \dots \cdot l_k$  experiments
- The easiest factorial design is the  $2^k$  with  $l_i = 2 \ \forall i = 1, \dots, k$ .

# A Problem

- *Full factorial design with replication*
  - ▣ Measure system response with all possible input combinations
  - ▣ Replicate each measurement  $n$  times to determine effect of measurement error
- $k$  factors,  $v$  levels,  $n$  replications
  - $n v^k$  experiments
- *Example:*
  - ▣  $k = 5$  input factors,  $v = 4$  levels,  $n = 3$
  - ▣ →  $3(4^5) = 3,072$  executions!

# Factorial Designs: $n2^k$ Experiments

- Special case of generalized  $m$ -factor experiments
- Restrict each factor to two possible values
  - ▣ High, low
  - ▣ On, off
- Find factors that have largest impact
- Full factorial design with only those factors

# Finding Sum of Squares Terms

| Sum of n measurements with (A,B) = (High, Low) | Factor A | Factor B |
|------------------------------------------------|----------|----------|
| yAB                                            | High     | High     |
| yAb                                            | High     | Low      |
| yaB                                            | Low      | High     |
| yab                                            | Low      | Low      |

# Contrasts for n2k with $k = 2$ factors

- $W_a = y_{AB} + y_{Ab} - y_{aB} - y_{ab}$
- $W_b = y_{AB} - y_{Ab} + y_{aB} - y_{ab}$
- $W_{ab} = y_{AB} - y_{Ab} - y_{aB} + y_{ab}$

| Measurements | Contrast |       |          |
|--------------|----------|-------|----------|
|              | $W_a$    | $W_b$ | $W_{ab}$ |
| $y_{AB}$     | +        | +     | +        |
| $y_{Ab}$     | +        | -     | -        |
| $y_{aB}$     | -        | +     | -        |
| $y_{ab}$     | -        | -     | +        |



# Contrasts for n2k with k = 3 factors

□  $W_{AC} = y_{abc} - y_{Abc} + y_{aBc} - y_{abC} - y_{ABc} + y_{AbC} - y_{aBC} + y_{ABC}$

| Means     | Contrast |       |       |          |          |          |           |
|-----------|----------|-------|-------|----------|----------|----------|-----------|
|           | $W_a$    | $W_b$ | $W_c$ | $W_{ab}$ | $W_{ac}$ | $W_{bc}$ | $W_{abc}$ |
| $y_{abc}$ | -        | -     | -     | +        | +        | +        | -         |
| $y_{Abc}$ | +        | -     | -     | -        | -        | +        | +         |
| $y_{aBc}$ | -        | +     | -     | -        | +        | -        | +         |
| ...       | ...      | ...   | ...   | ...      | ...      | ...      | ...       |

## $n2^k$ with $k = 3$ factors

- $df(\text{each effect}) = 1$ , since only two levels measured
- $SST = SSA + SSB + SSC + SSAB + SSAC + SSBC + SSABC$
- $df(SSE) = (n-1)2^3$
- Then perform ANOVA as before
- Easily? generalizes to  $m > 3$  factors
- $SSAC = \frac{w_{AC}^2}{n2^3}$

# $2^k$ Matrix

| Experiment | Factor 1 | Factor 2 | .... | Factor k | Answer |
|------------|----------|----------|------|----------|--------|
| 1          | -        | -        |      | -        | R1     |
| 2          | +        | -        |      | -        | R2     |
| 3          | -        | -        |      | -        | R3     |
| 4          | +        | -        |      | -        | R4     |
| 5          | -        | +        |      | -        | R5     |
| 6          | +        | +        |      | -        | R6     |
| $2^k$      | +        | +        |      | +        | $R2^k$ |

# Yates notation

- Levels: High (+), Low ("-") for factors A and B

| A | B | Yates notation | Meaning                   |
|---|---|----------------|---------------------------|
| - | - | (1)            | Both factors on low level |
| + | - | a              | A in +, B in -            |
| - | + | b              | A in - B in +             |
| + | + | ab             | A in + B in +             |

# Important Points

- Experimental design is used to
  - ▣ Isolate the effects of each input variable.
  - ▣ Determine the effects of interactions.
  - ▣ Determine the magnitude of the error
  - ▣ Obtain maximum information for given effort
- Expand 1-factor ANOVA to  $k$  factors
- Use  $n2^k$  design to reduce the number of experiments needed
  - ▣ But loses some information
  - ▣ Useful to underline the tendency with economy of experiments.

# And the ANOVA table?

- $H_0: \mu_1 = \mu_2 = \mu_3 \dots = \mu_k$
- $H_1$ : Means are not all equal.
  - ▣ The number of points,  $N = 2^2n = 4n$  where  $n$  are the replications for each scenario.
  - ▣  $G$ =grand total., and  $(a)$ ,  $(b)$ ,  $(ab)$  contrast of the total

| Source of variation | Degree of freedom (df) | Sum of Squares (SS) | Mean Square $MSS=SS/df$ | F-ratio    |
|---------------------|------------------------|---------------------|-------------------------|------------|
| A                   | $(a-1)=2-1=1$          | $SS_A$              | $MSA$                   | $MSA/MSE$  |
| B                   | $(b-1)=2-1=1$          | $SS_B$              | $MSB$                   | $MSB/MSE$  |
| AB                  | $(a-1)(b-1)=1$         | $SS_{AB}$           | $MSAB$                  | $MSAB/MSE$ |
| Error               | $4n-4=4(n-1)$          | $SSE$               | $MSE$                   |            |
| Total               | $4n-1$                 | $SST$               |                         |            |

$$SST = \sum \sum x_{ij}^2 - \frac{G^2}{N}$$

$$SS_A = \frac{(a)^2}{2^2n}$$

$$SS_B = \frac{(b)^2}{2^2n}$$

$$SS_{AB} = \frac{(ab)^2}{2^2n}$$

# Exercise 1

- We have on a factory three different Machines:
  - ▣ A, with speed from 2 to 10
  - ▣ B, with speed of 2 and 3
  - ▣ C, with speed of 2 but that can be changed for other machine with a speed of 3 for 1000€.
- Define a table for an  $2^k$  experimental design that allows to analyze this.



# Solution

|   | A            | B          | C                  | Answer |
|---|--------------|------------|--------------------|--------|
| 1 | - (means 2)  | -(means 2) | - (means 2)        |        |
| 2 | -            | -          | + (means 3, 1000€) |        |
| 3 | -            | +(means 3) | -                  |        |
| 4 | -            | +          | +                  |        |
| 5 | + (means 10) | -          | -                  |        |
| 6 | +            | -          | +                  |        |
| 7 | +            | +          | -                  |        |
| 8 | +            | +          | +                  |        |



# Yates algorithm

Simplifying the interaction calculus on a  $2^k$  factorial design

# $2^k$ factorial designs

## Advantages

- Determination of the tendency with experiments economy (smoothness).
- Possibility to evolve to composite designs (local exploration).
- Basis for factorial fractional designs (rapid vision of multiple factors).
- Easy analysis and interpretation.

# $2^k$ Matrix example

| Experiment | A | B | C | Answer |
|------------|---|---|---|--------|
| 1          | - | - | - | 60     |
| 2          | + | - | - | 72     |
| 3          | - | + | - | 54     |
| 4          | + | + | - | 68     |
| 5          | - | - | + | 52     |
| 6          | + | - | + | 83     |
| 7          | - | + | + | 45     |
| 8          | + | + | + | 80     |

# Effects calculus example

- *Effect*  $A = \frac{A_1B_0 - A_1B_1}{2} - \frac{A_0B_0 - A_0B_1}{2}$
- *Effect*  $B = \frac{A_1B_1 - A_0B_1}{2} - \frac{A_0B_0 - A_1B_0}{2}$
- $\text{contrast} = \bar{y}_+ - \bar{y}_-$

$$A = \frac{72 + 68 + 83 + 80}{4} - \frac{60 + 54 + 52 + 45}{4} = 23$$

$$B = \frac{54 + 68 + 45 + 80}{4} - \frac{60 + 72 + 52 + 83}{4} = -5$$

$$C = \frac{52 + 83 + 45 + 80}{4} - \frac{60 + 72 + 54 + 68}{4} = 1.5$$

$$AC = \frac{y_1 + y_3 + y_6 + y_8}{4} - \frac{y_2 + y_4 + y_5 + y_7}{4} = 10$$

$$ABC = \frac{y_{21} + y_3 + y_5 + y_8}{4} - \frac{y_1 + y_4 + y_6 + y_7}{4} = 0.5$$

# Frank Yates



- A pioneer of the Operation research of the s.XX.

# Yates algorithm

To make systematic the interactions calculus using a table.

- Add the **answer** in the column “i” in the standard form of the matrix of the experimental design.
- Add **auxiliary columns** as factors exists.
- Add a new column dividing the first value of the last auxiliary column by the number of scenarios “E”, and the others by the half of “E”.

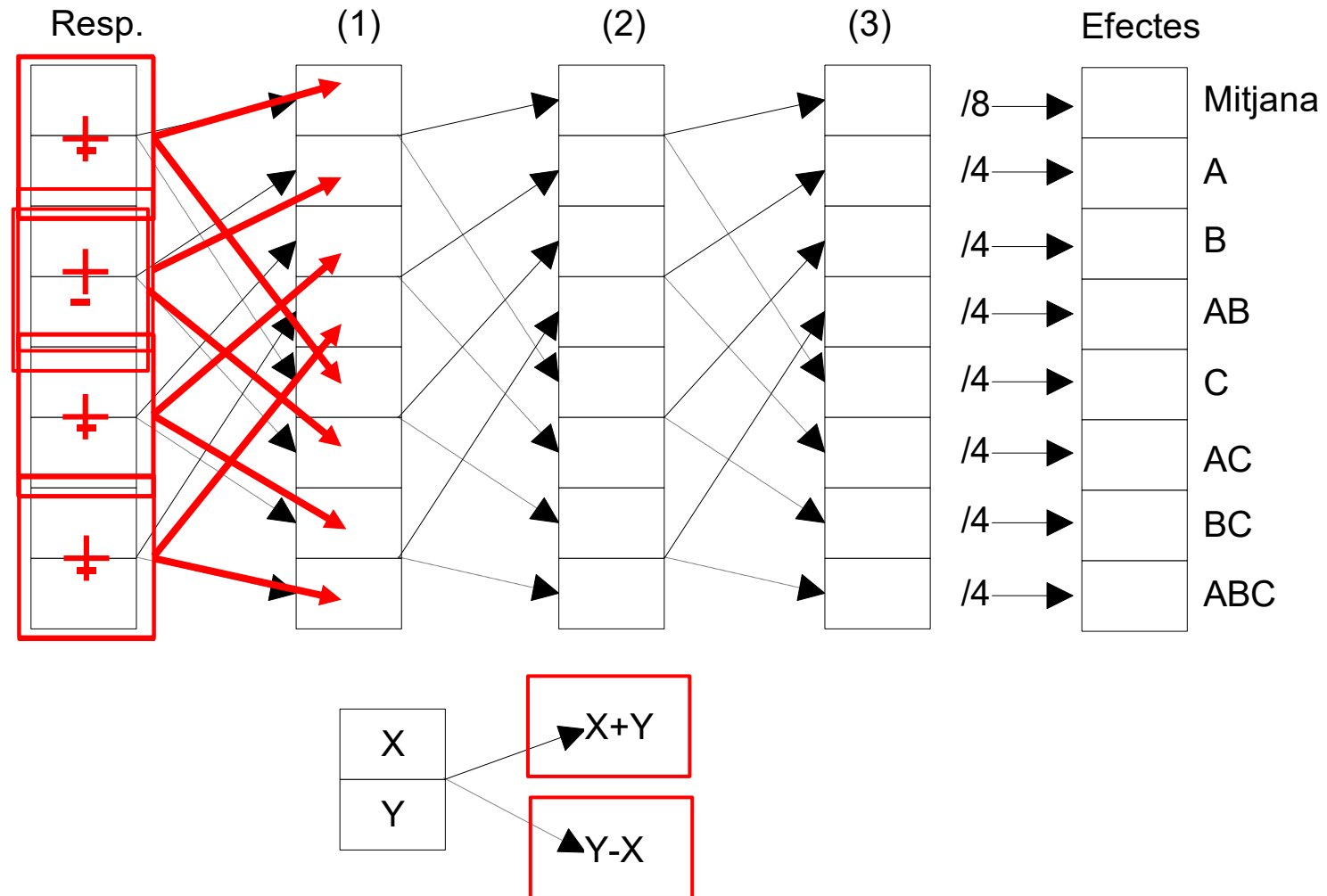
|   |   |   |
|---|---|---|
| - | - | - |
| + | - | - |
| - | + | - |
| + | + | - |
| - | - | + |
| + | - | + |
| - | + | + |
| + | + | + |

standard form

# Yates algorithm

- In the last column the first value is the mean of the answers, the last values are the effects.
- The correspondence between the values and effects is done through localize the + values in the corresponding rows of the matrix. A value with a single + in the B column is representing the principal effect of B. A row with two + on A and C corresponds to the interaction of AC, etc.

# Yates algorithm

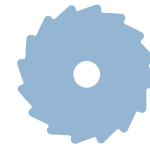




# Yates algorithm example

| Exp. | A | B | C | Answer | (1) | (2) | (3) | div. | effect | Id   |
|------|---|---|---|--------|-----|-----|-----|------|--------|------|
| 1    | - | - | - | 60     | 132 | 254 | 514 | 8    | 64.25  | Mean |
| 2    | + | - | - | 72     | 122 | 260 | 92  | 4    | 23.0   | A    |
| 3    | - | + | - | 54     | 135 | 26  | -20 | 4    | -5.0   | B    |
| 4    | + | + | - | 68     | 125 | 66  | 6   | 4    | 1.5    | AB   |
| 5    | - | - | + | 52     | 12  | -10 | 6   | 4    | 1.5    | C    |
| 6    | + | - | + | 83     | 14  | -10 | 40  | 4    | 10.0   | AC   |
| 7    | - | + | + | 45     | 31  | 2   | 0   | 4    | 0.0    | BC   |
| 8    | + | + | + | 80     | 35  | 4   | 2   | 4    | 0.5    | ABC  |

# Wooden industry example



**Wooden industry that allows to reduce the cost.**

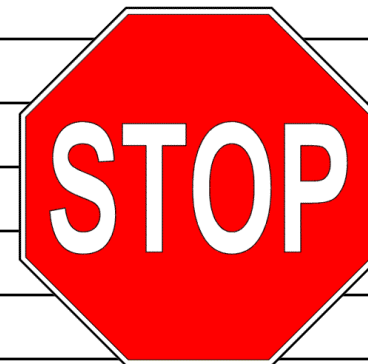


**4 variables to consider**

- A: Change the light to natural light (open the ceiling).
- B: Increase the speed of the machines.
- C: Increase the lubricant use.
- D: Increase the working space.

# Wooden industry example

| Comb. | 1 | 2 | 3 | 4 | Description                        | obs. |
|-------|---|---|---|---|------------------------------------|------|
| (1)   | - | - | - | - |                                    | 71   |
| a     | + | - | - | - | Natural light                      | 61   |
| b     | - | + | - | - | Increase the speed of the machines | 90   |
| ab    | + | + | - | - |                                    | 82   |
| c     | - | - | + | - | Increase the use of lubricant      | 68   |
| ac    | + | - | + | - |                                    | 61   |
| bc    | - | + | + | - |                                    | 87   |
| abc   | + | + | + | - |                                    | 80   |
| d     | - | - | - | + | Increase the working space.        | 61   |
| ad    | + | - | - | + |                                    | 50   |
| bd    | - | + | - | + |                                    | 89   |
| abd   | + | + | - | + |                                    | 83   |
| cd    | - | - | + | + |                                    | 59   |
| acd   | + | - | + | + |                                    | 51   |
| bcd   | - | + | + | + |                                    | 85   |
| abcd  | + | + | + | + |                                    | 78   |



# Wooden industry example

| Comb. | obs. | 1 | 2 | 3 | 4 | Effect | Description |
|-------|------|---|---|---|---|--------|-------------|
| (1)   | 71   |   |   |   |   |        |             |
| a     | 61   |   |   |   |   |        |             |
| b     | 90   |   |   |   |   |        |             |
| ab    | 82   |   |   |   |   |        |             |
| c     | 68   |   |   |   |   |        |             |
| ac    | 61   |   |   |   |   |        |             |
| bc    | 87   |   |   |   |   |        |             |
| abc   | 80   |   |   |   |   |        |             |
| d     | 61   |   |   |   |   |        |             |
| ad    | 50   |   |   |   |   |        |             |
| bd    | 89   |   |   |   |   |        |             |
| abd   | 83   |   |   |   |   |        |             |
| cd    | 59   |   |   |   |   |        |             |
| acd   | 51   |   |   |   |   |        |             |
| bcd   | 85   |   |   |   |   |        |             |
| abcd  | 78   |   |   |   |   |        |             |

# Wooden industry example

| Comb. | obs. | 1   | 2   | 3   | 4    | Effect | Description |
|-------|------|-----|-----|-----|------|--------|-------------|
| (1)   | 71   | 132 | 304 | 600 | 1156 | 72,25  | Mean        |
| a     | 61   | 172 | 296 | 556 | -64  | -8     | A           |
| b     | 90   | 129 | 283 | -32 | 192  | 24     | B           |
| ab    | 82   | 167 | 273 | -32 | 8    | 1      | AB          |
| c     | 68   | 111 | -18 | 78  | -18  | -2,25  | C           |
| ac    | 61   | 172 | -14 | 114 | 6    | 0,75   | AC          |
| bc    | 87   | 110 | -17 | 2   | -10  | -1,25  | BC          |
| abc   | 80   | 163 | -15 | 6   | -6   | -0,75  | ABC         |
| d     | 61   | -10 | 40  | -8  | -44  | -5,5   | D           |
| ad    | 50   | -8  | 38  | -10 | 0    | 0      | AD          |
| bd    | 89   | -7  | 61  | 4   | 36   | 4,5    | BD          |
| abd   | 83   | -7  | 53  | 2   | 4    | 0,5    | ABD         |
| cd    | 59   | -11 | 2   | -2  | -2   | -0,25  | CD          |
| acd   | 51   | -6  | 0   | -8  | -2   | -0,25  | ACD         |
| bcd   | 85   | -8  | 5   | -2  | -6   | -0,75  | BCD         |
| abcd  | 78   | -7  | 1   | -4  | -2   | -0,25  | ABCD        |

## Exercise 2: Clean industry

- The system processing time follows an **exponential** distribution ( $r = 1 - e^{-\alpha \cdot x} \Rightarrow x = \frac{\ln(1-r)}{-\alpha} = \frac{\ln(r)}{-\alpha}$ ) with parameter  $\alpha = 1/\mu$ . The value of  $\mu$  is determined by three additive factors:
  - ▣ Cleaning Time: A variable duration ranging from 10 to 50 seconds.
  - ▣ Gluing Capacity: Based on a fleet of 1 to 5 machines. Each machine added beyond the second reduces the total time by 1 second.
  - ▣ Post-Processing Staff: Utilizing 1 worker adds 1 second, while 2 workers reduce this contribution to 0.5 seconds.

# Answer

| Cleaner | Machines | Workers | VALUES |    | $\mu$ | $1/\mu$ | x1       | x2       | x3       | mean     |          |          |          |          |          |                          |
|---------|----------|---------|--------|----|-------|---------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--------------------------|
| -       | -        | -       | 50     | 0  | 1     | 51      | 0,019608 | 56,1716  | 34,7366  | 111,6928 | 67,53366 | 89,94953 | 108,0193 | 163,0003 | 20,37504 | Mean                     |
| +       | -        | -       | 10     | 0  | 1     | 11      | 0,090909 | 0,774709 | 6,738894 | 59,73398 | 22,41586 | 18,06978 | 54,981   | -82,0152 | -20,5038 | Cleaner                  |
| -       | +        | -       | 50     | -4 | 1     | 47      | 0,021277 | 22,08345 | 20,38672 | 1,314331 | 14,59483 | 29,67531 | -56,2377 | -76,2494 | -19,0623 | Machines                 |
| +       | +        | -       | 10     | -4 | 1     | 7       | 0,142857 | 1,866081 | 7,774161 | 0,784596 | 3,474946 | 25,30569 | -25,7775 | 48,66833 | 12,16708 | Cleaner*Machines         |
| -       | -        | +       | 50     | 0  | 0,5   | 50,5    | 0,019802 | 11,60592 | 58,552   | 4,691006 | 24,94964 | -45,1178 | -71,8797 | -53,0383 | -13,2596 | Workers                  |
| +       | -        | +       | 10     | 0  | 0,5   | 10,5    | 0,095238 | 6,116605 | 2,435266 | 5,625142 | 4,725671 | -11,1199 | -4,36963 | 30,46016 | 7,615041 | Cleaner*Workers          |
| -       | +        | +       | 50     | -4 | 0,5   | 46,5    | 0,021505 | 27,90595 | 7,941664 | 10,44125 | 15,42962 | -20,224  | 33,99791 | 67,51012 | 16,87753 | Machines*Workers         |
| +       | +        | +       | 10     | -4 | 0,5   | 6,5     | 0,153846 | 22,2303  | 1,848943 | 5,548947 | 9,876064 | -5,55356 | 14,67041 | -19,3275 | -4,83188 | Cleaner*Machines*Workers |

$$r = 1 - e^{-\alpha \cdot x} \Rightarrow x = \frac{\ln(1-r)}{-\alpha} = \frac{\ln(r)}{-\alpha}$$



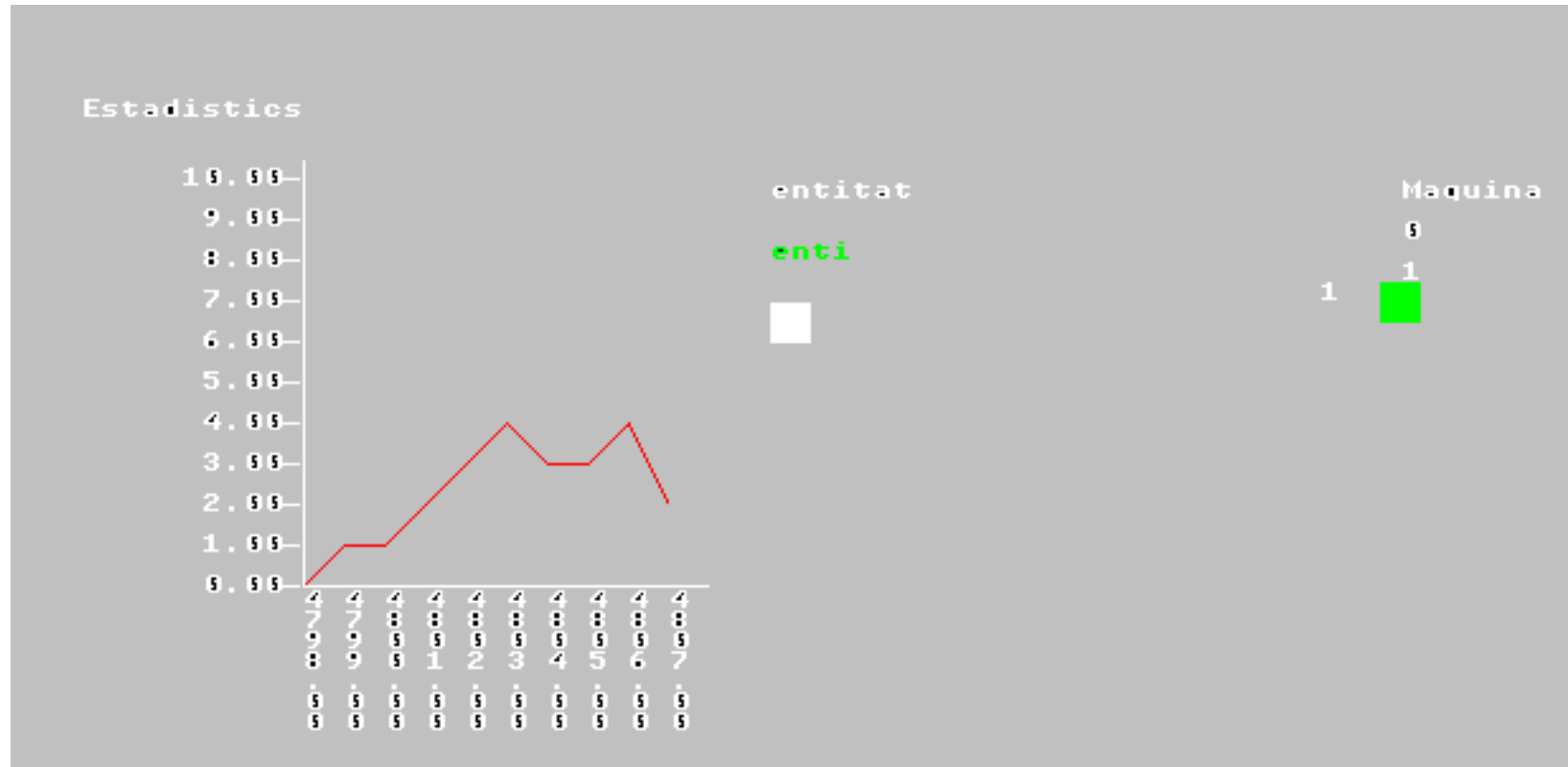
# Replications

Number of replications calculus.

Methods to perform the replications.



# Interest variable calculus



# Experimentation

- Be  $x$  an interest variable

$x_{11}, \dots, x_{1i}, \dots, x_{1m}$

$x_{21}, \dots, x_{2i}, \dots, x_{2m}$

.....

$x_{n1}, \dots, x_{ni}, \dots, x_{nm}$

- $n$  is the number of replications.
- $x_i$  is the value of each one of the replications.

# Sample mean and variance

$$\overline{X} = \frac{\sum_{i=1}^n x_i}{n}$$

$$S^2 = \frac{\sum_{i=1}^n x_i - \overline{X}}{n - 1}$$

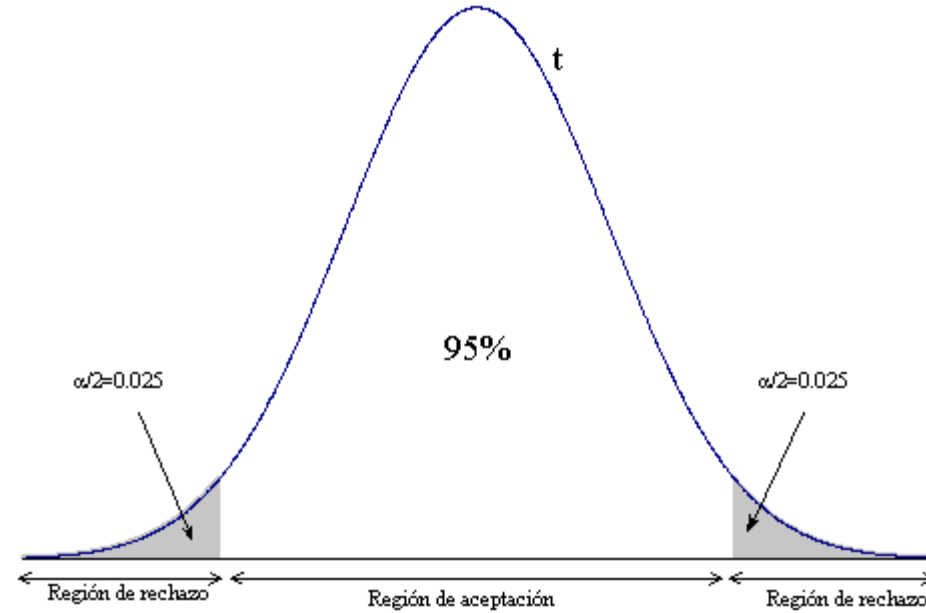
The use of the term  $n - 1$  is called Bessel's correction, for the *unbiased sample variance*

# Confidence interval

- Need to know how far is  $\mu$  and  $\bar{y}$ .
- Student's t-distribution of  $n-1$  degrees of freedom.

$$\bar{y} \pm t_{1-\alpha/2, n-1} \sqrt{\frac{S^2}{n}}$$

# Student's t-distribution



# What is the correct n?

| Replication | Value from the model |
|-------------|----------------------|
| 1           | 28.841               |
| 2           | 35.965               |
| 3           | 31.219               |
| 4           | 37.090               |
| 5           | 38.734               |
| 6           | 30.923               |
| 7           | 30.443               |
| 8           | 32.175               |
| 9           | 30.683               |
| 10          | 28.745               |

# Calculus of S on X

---

$$\bar{y} = 32.4818$$

$$S^2 = 3.5149$$

# Calculus of the self-confidence interval

$$h = t_{1-\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

$$t_{9, 0.975} = 2,26$$
$$h = 2,512$$

|    | 1 - α |       |       |       |       |              |        |        |
|----|-------|-------|-------|-------|-------|--------------|--------|--------|
| r  | 0.75  | 0.80  | 0.85  | 0.90  | 0.95  | 0.975        | 0.99   | 0.995  |
| 1  | 1.000 | 1.376 | 1.963 | 3.078 | 6.314 | 12.706       | 31.821 | 63.657 |
| 2  | 0.816 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303        | 6.965  | 9.925  |
| 3  | 0.765 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182        | 4.541  | 5.841  |
| 4  | 0.741 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776        | 3.747  | 4.604  |
| 5  | 0.727 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571        | 3.365  | 4.032  |
| 6  | 0.718 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447        | 3.143  | 3.707  |
| 7  | 0.711 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365        | 2.998  | 3.499  |
| 8  | 0.706 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306        | 2.896  | 3.355  |
| 9  | 0.703 | 0.883 | 1.100 | 1.383 | 1.833 | <b>2.262</b> | 2.821  | 3.250  |
| 10 | 0.700 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228        | 2.764  | 3.169  |



## Confidence interval:

- (  $32.4818 - 2.512 = 29.9698$ ,  $32.4818 + 2.512 = 34.9938$  )
- The interpretation is that with a probability of 0.95, the random interval (29.9698, 34.9938) includes the real value of the mean.

## More replications needed.

- If we specify that we want an interval between a 5% of the sample mean with a confidence level of a 95%, we need more replications.
- $0.05 \cdot (32.4818) = 1.62$  but we have 2.512

# Number of needed replications

- where:
- $n$  = initial number of replications.
- $n^*$  = total replications needed.
- $h$  = half-range of the confidence interval for the initial number of replications.
- $h^*$  = half-range of the confidence interval for all the replications (the desired half-range).

$$n^* = n \left( \frac{h}{h^*} \right)^2$$

# Number of replications calculus.

---

$$n^* = 10 \left( \frac{2.512}{1.62} \right)^2 = 24.04$$

# More replications...

| Replication | Value from the model |
|-------------|----------------------|
| 11          | 33.020               |
| 12          | 29.472               |
| 13          | 27.693               |
| 14          | 31.803               |
| 15          | 30.604               |
| 16          | 33.227               |
| 17          | 28.085               |
| 18          | 35.910               |
| 19          | 30.729               |
| 20          | 30.844               |
| 21          | 32.420               |
| 22          | 39.040               |
| 23          | 32.341               |
| 24          | 34.310               |
| 25          | 28.418               |

## New mean and variance

---

$$\bar{y} = 32.1094$$

$$S^2 = 3.1903$$

# New self-confidence interval

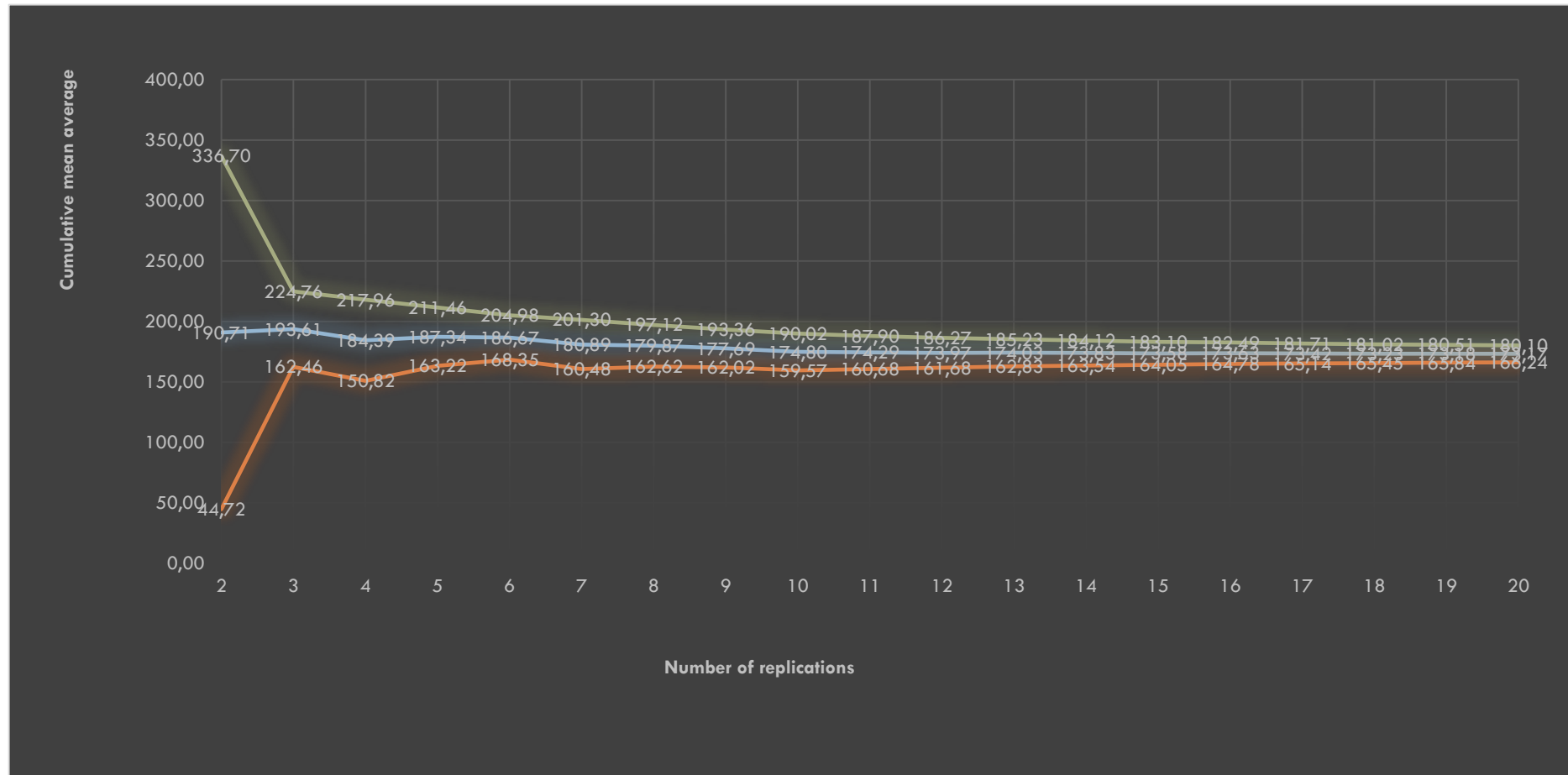
- In that case is enough, but the process can be iterative.

$$h = t_{1-\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

$$h = 1.3144 < 1.62$$

$$n^* = n \left( \frac{h}{h^*} \right)^2$$

# With more replications





# Exercise 3

- Calculate the amount of replications needed for the Exercise 2

| Cleaner | Machines | Workers | x1     | x2     |
|---------|----------|---------|--------|--------|
| -       | -        | -       | 20,885 | 20,261 |
| -       | -        | +       | 33,836 | 36,368 |
| -       | +        | -       | 9,9099 | 142    |
| -       | +        | +       | 17,766 | 131,13 |
| +       | -        | -       | 42,759 | 0,0402 |
| +       | -        | +       | 5,7025 | 2,327  |
| +       | +        | -       | 10,481 | 8,7404 |
| +       | +        | +       | 5,9775 | 5,1167 |

|     | 1 - $\alpha$ |       |       |       |       |        |        |        |
|-----|--------------|-------|-------|-------|-------|--------|--------|--------|
| r   | 0.75         | 0.80  | 0.85  | 0.90  | 0.95  | 0.975  | 0.99   | 0.995  |
| 1   | 1.000        | 1.376 | 1.963 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 |
| 2   | 0.816        | 1.061 | 1.386 | 1.886 | 2.920 | 4.303  | 6.965  | 9.925  |
| 3   | 0.765        | 0.978 | 1.250 | 1.638 | 2.353 | 3.182  | 4.541  | 5.841  |
| 4   | 0.741        | 0.941 | 1.190 | 1.533 | 2.132 | 2.776  | 3.747  | 4.604  |
| 5   | 0.727        | 0.920 | 1.156 | 1.476 | 2.015 | 2.571  | 3.365  | 4.032  |
| 6   | 0.718        | 0.906 | 1.134 | 1.440 | 1.943 | 2.447  | 3.143  | 3.707  |
| 7   | 0.711        | 0.896 | 1.119 | 1.415 | 1.895 | 2.365  | 2.998  | 3.499  |
| 8   | 0.706        | 0.889 | 1.108 | 1.397 | 1.860 | 2.306  | 2.896  | 3.355  |
| 9   | 0.703        | 0.883 | 1.100 | 1.383 | 1.833 | 2.262  | 2.821  | 3.250  |
| 10  | 0.700        | 0.879 | 1.093 | 1.372 | 1.812 | 2.228  | 2.764  | 3.169  |
| 11  | 0.697        | 0.876 | 1.088 | 1.363 | 1.796 | 2.201  | 2.718  | 3.106  |
| 12  | 0.695        | 0.873 | 1.083 | 1.356 | 1.782 | 2.179  | 2.681  | 3.055  |
| 13  | 0.694        | 0.870 | 1.079 | 1.350 | 1.771 | 2.160  | 2.650  | 3.012  |
| 14  | 0.692        | 0.868 | 1.076 | 1.345 | 1.761 | 2.145  | 2.624  | 2.977  |
| 15  | 0.691        | 0.866 | 1.074 | 1.341 | 1.753 | 2.131  | 2.602  | 2.947  |
| 16  | 0.690        | 0.865 | 1.071 | 1.337 | 1.746 | 2.120  | 2.583  | 2.921  |
| 17  | 0.689        | 0.863 | 1.069 | 1.333 | 1.740 | 2.110  | 2.567  | 2.898  |
| 18  | 0.688        | 0.862 | 1.067 | 1.330 | 1.734 | 2.101  | 2.552  | 2.878  |
| 19  | 0.688        | 0.861 | 1.066 | 1.328 | 1.729 | 2.093  | 2.539  | 2.861  |
| 20  | 0.687        | 0.860 | 1.064 | 1.325 | 1.725 | 2.086  | 2.528  | 2.845  |
| 21  | 0.686        | 0.859 | 1.063 | 1.323 | 1.721 | 2.080  | 2.518  | 2.831  |
| 22  | 0.686        | 0.858 | 1.061 | 1.321 | 1.717 | 2.074  | 2.508  | 2.819  |
| 23  | 0.685        | 0.858 | 1.060 | 1.319 | 1.714 | 2.069  | 2.500  | 2.807  |
| 24  | 0.685        | 0.857 | 1.059 | 1.318 | 1.711 | 2.064  | 2.492  | 2.797  |
| 25  | 0.684        | 0.856 | 1.058 | 1.316 | 1.708 | 2.060  | 2.485  | 2.787  |
| 26  | 0.684        | 0.856 | 1.058 | 1.315 | 1.706 | 2.056  | 2.479  | 2.779  |
| 27  | 0.684        | 0.855 | 1.057 | 1.314 | 1.703 | 2.052  | 2.473  | 2.771  |
| 28  | 0.683        | 0.855 | 1.056 | 1.313 | 1.701 | 2.048  | 2.467  | 2.763  |
| 29  | 0.683        | 0.854 | 1.055 | 1.311 | 1.699 | 2.045  | 2.462  | 2.756  |
| 30  | 0.683        | 0.854 | 1.055 | 1.310 | 1.697 | 2.042  | 2.457  | 2.750  |
| 40  | 0.681        | 0.851 | 1.050 | 1.303 | 1.684 | 2.021  | 2.423  | 2.704  |
| 60  | 0.679        | 0.848 | 1.046 | 1.296 | 1.671 | 2.000  | 2.390  | 2.660  |
| 120 | 0.677        | 0.845 | 1.041 | 1.289 | 1.658 | 1.980  | 2.358  | 2.617  |
| ∞   | 0.674        | 0.842 | 1.036 | 1.282 | 1.645 | 1.960  | 2.326  | 2.576  |

# Replications

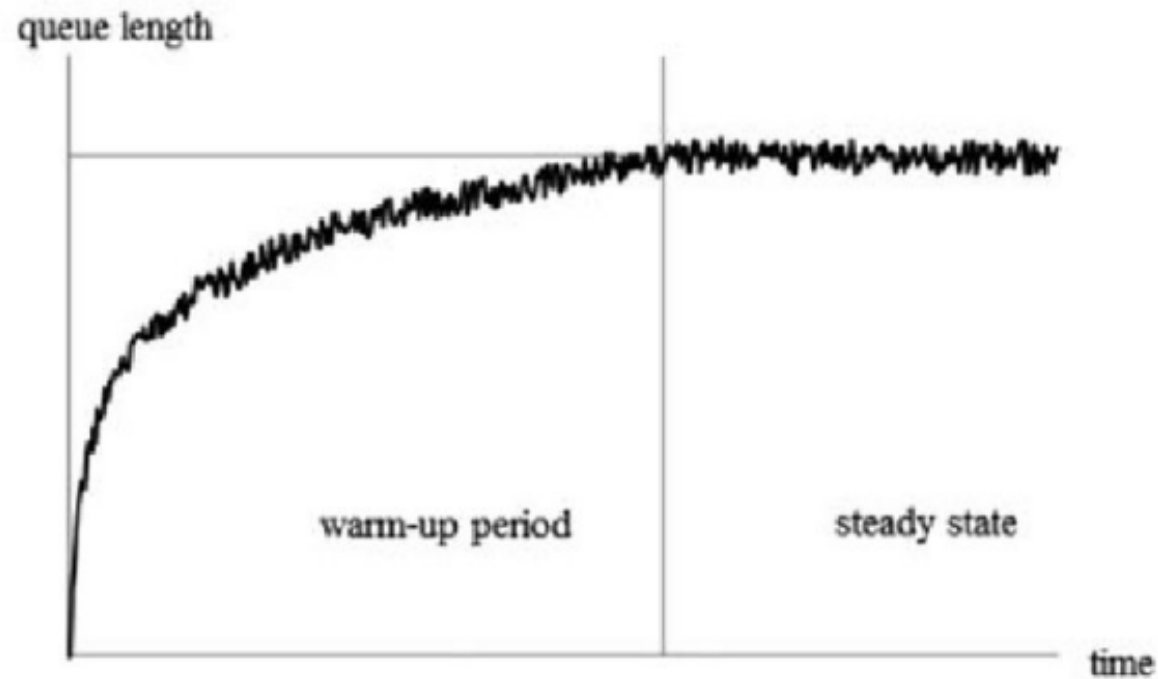
Methods to execute the replications.

# Kind of simulations

---

- **Terminate simulations (for terminate systems):** Simulations where a condition defines the end of the execution. Usually time.
- **Non terminate simulations (for non terminate systems):** Simulations without this condition.

# Transient period

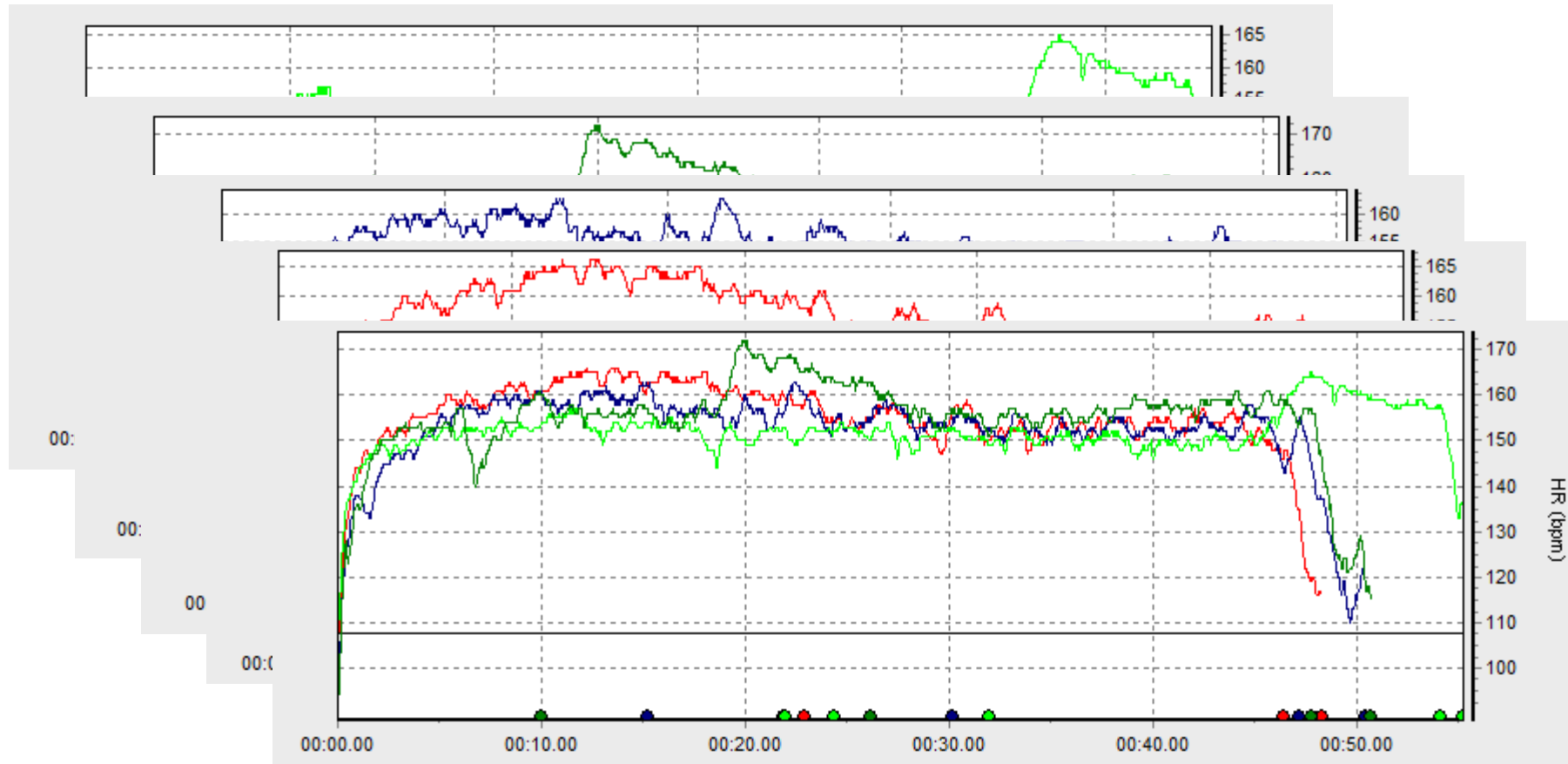


G. Chen and Z. Z. Yang, Methods for estimating vehicle queues at a marine terminal: A computational comparison. *Int. J. Appl. Math. Comput. Sci.* **24**, 611 (2014).

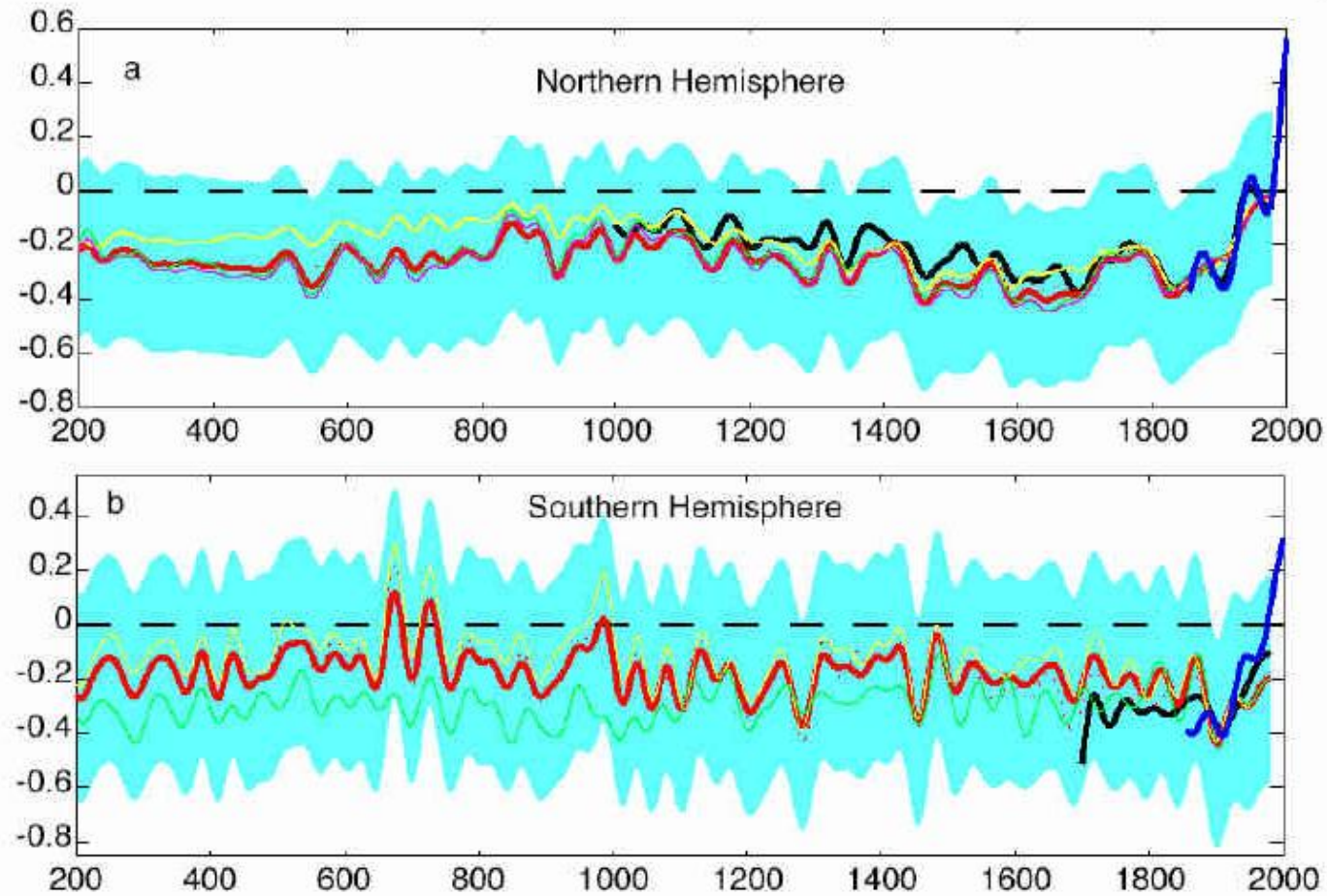
# Independent repetitions

- From the same initial state of the model, that means, with the same parameterizations and behavior, only random numbers to be used on the GAV are changed.
- This different RNG allows test again and again the new system with the different possible values of the variables that are not controlled (random variables).

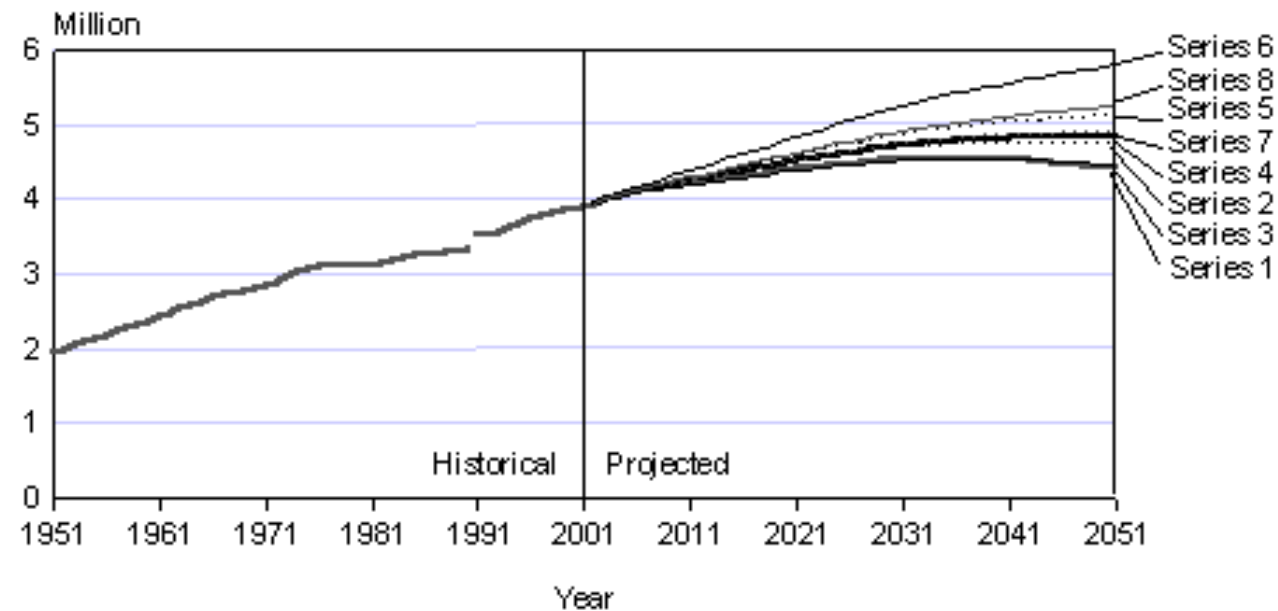
# Independent repetitions



# Independent repetitions



# Independent repetitions

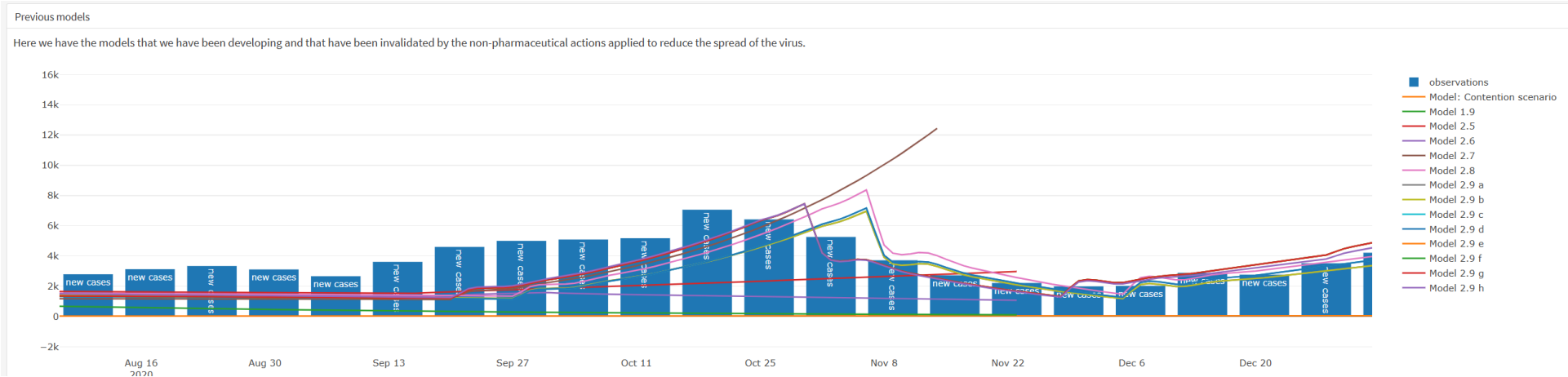
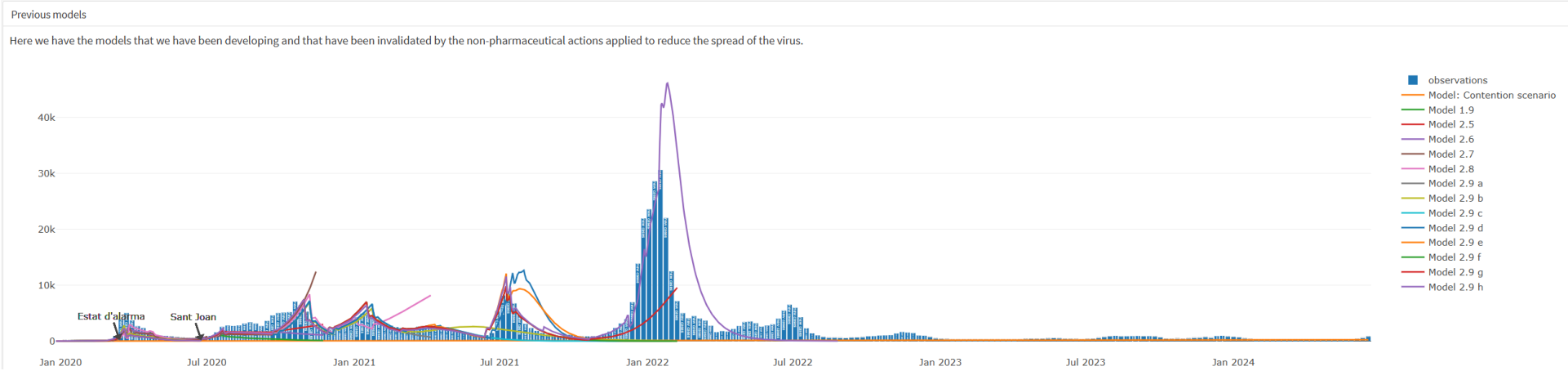


**Note** The break in series between 1990 and 1991 denotes a change from the de facto population concept to the resident population concept.



# Independent repetitions

SDL-PAND KPIs Model actual



# Transient elimination

## □ Definitions:

- $k$ : number of deleted observations.
- $m$ : amount of observations in a single run.
- $n$ : amount of runs (replications).

# Transient elimination

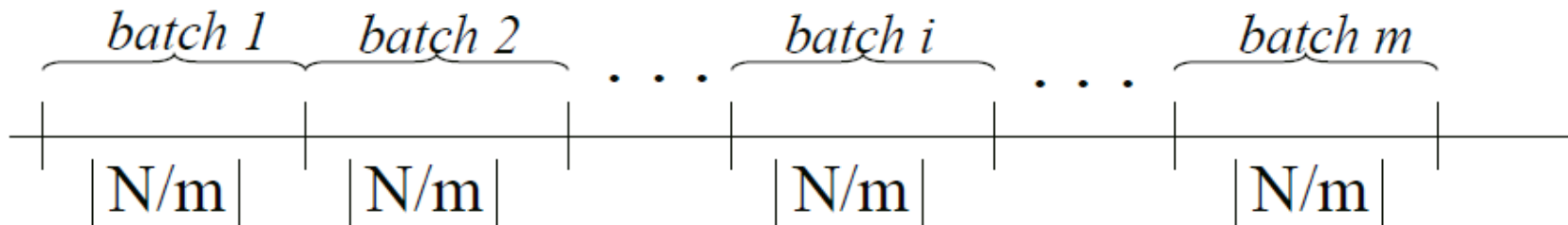
- Step 1: Compute the average of the  $j$  observation's over all runs.
  - ▣  $\bar{x}_j = \frac{1}{m} \sum_{i=1}^m x_{i,j}$
- Step 2: Compute the overall average.
  - ▣  $\bar{\bar{x}} = \frac{1}{n} \sum_{j=1}^n \bar{x}_j$
- Step 3:  $K=1$

# Transient elimination

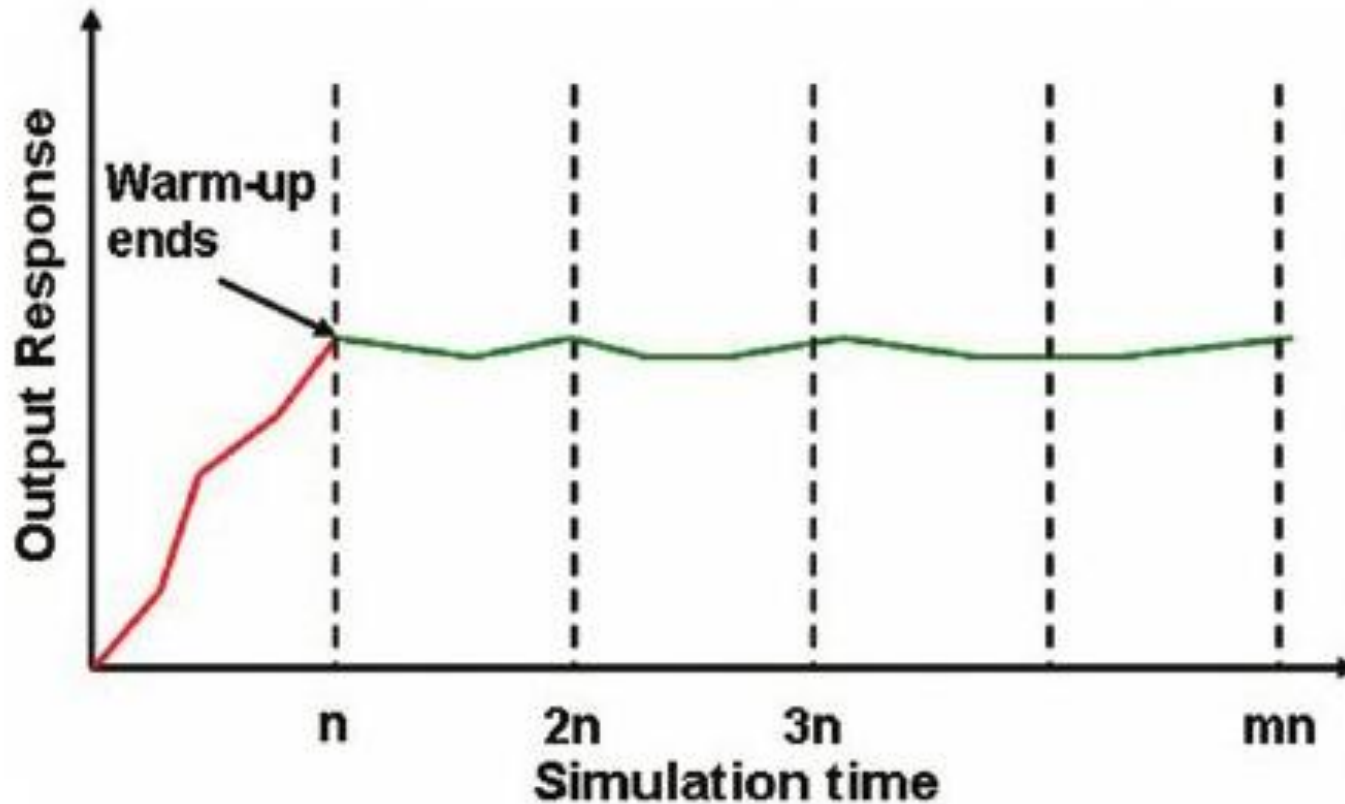
- Step 4: Compute the overall average of the  $j$  observation's over all runs without the first  $k$  observations.
  - ▣  $\bar{\bar{x}}_k = \frac{1}{n-k} \sum_{j=k+1}^n \bar{x}_j$
- Step 5: Calculate the relative change
  - ▣  $\Delta = \frac{\bar{\bar{x}}_k - \bar{\bar{x}}}{\bar{\bar{x}}}$
- If  $|\Delta_k - \Delta_{k-1}| > \text{threshold}$  then increment  $k$  and go to step 4, else remove  $k$  observations and use

# Batch means

- Execute a long simulation and then divide it in different blocks, or execution bags “batches”.
  - ▣ We work with the mean values of these observations.
  - ▣ The size of the “batches” is  $n = \lfloor N/m \rfloor$
- Each one of these observations are considered as independent.
- Is desirable to determine what must be the required long of each one of these execution blocks, to assure the correctness of the experiment.

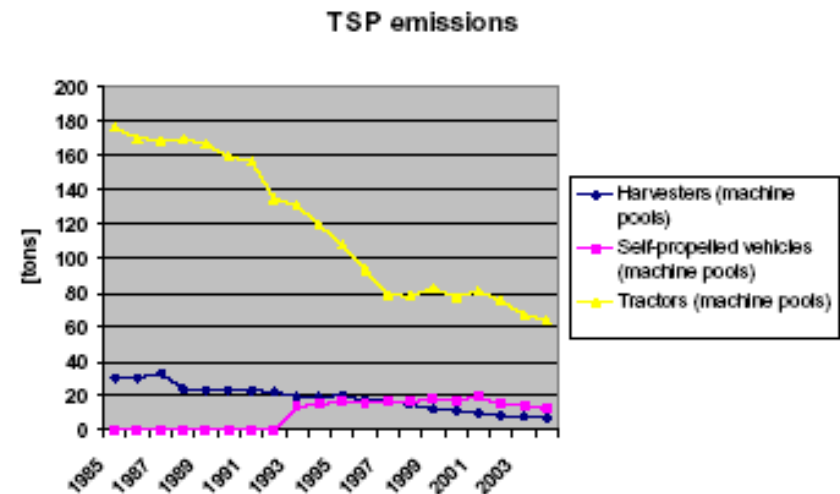
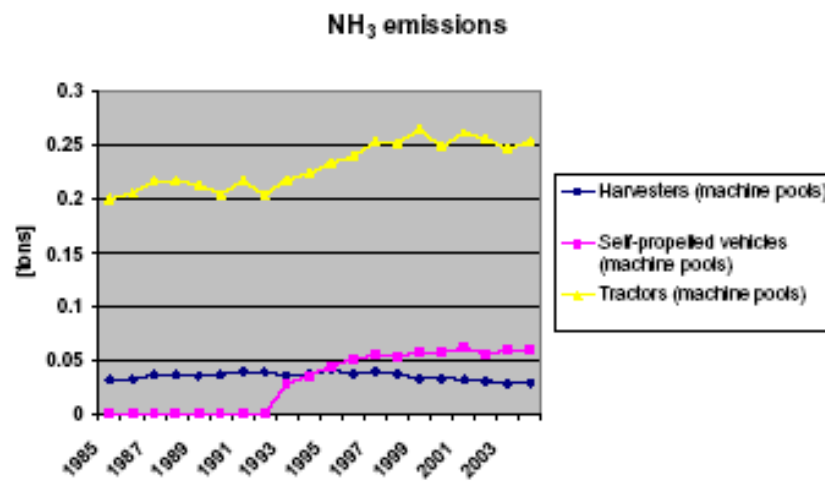
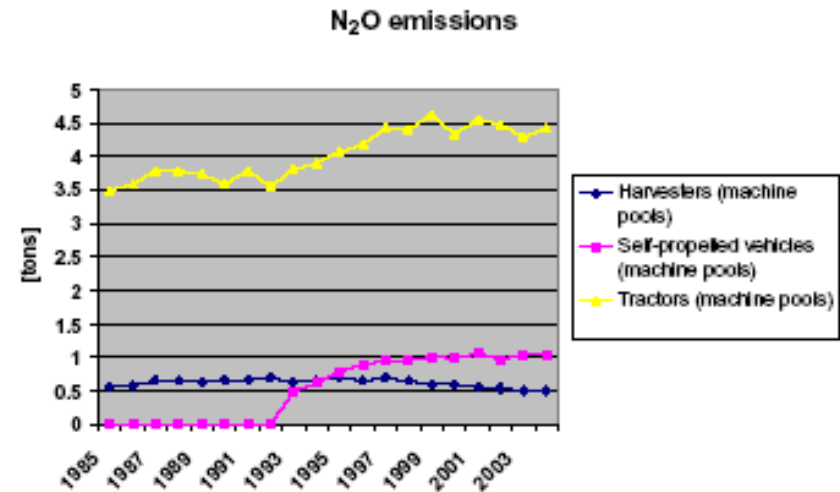
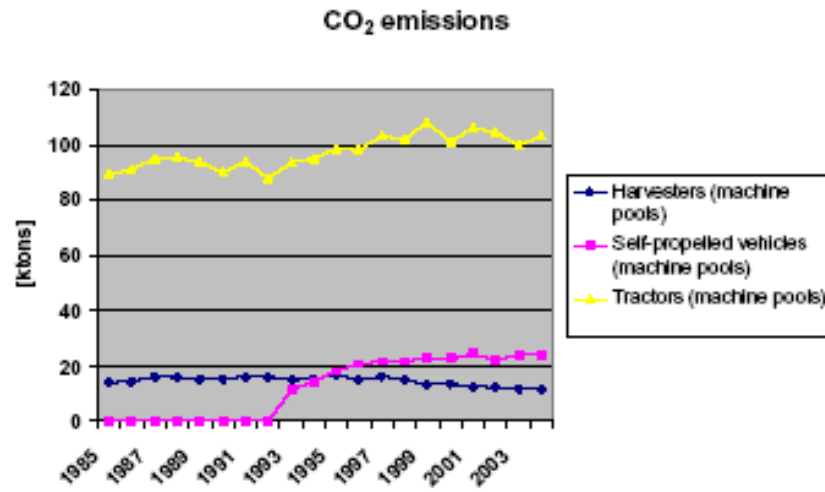


# Batch means



G. Chen and Z. Z. Yang, A review of techniques for the analysis of simulation output, Int. J. Appl. Math. Comput. Sci. **24**, 611 (2014).

# Batch means



# Transient elimination

- Step 1: Set  $n=2$
- Step 2: Compute the average of the “i” batch.
  - ▣  $\bar{x}_i = \frac{1}{n} \sum_{j=1}^n x_{i,j}$
- Step 3: Compute the overall average.
  - ▣  $\bar{\bar{x}} = \frac{1}{m} \sum_{i=1}^m \bar{x}_i$
- Step 4: Compute the variance of the batch means.
  - ▣  $Var(\bar{x}) = \frac{1}{m-1} \sum_{i=1}^m (\bar{x}_i - \bar{\bar{x}})^2$

Definitions:

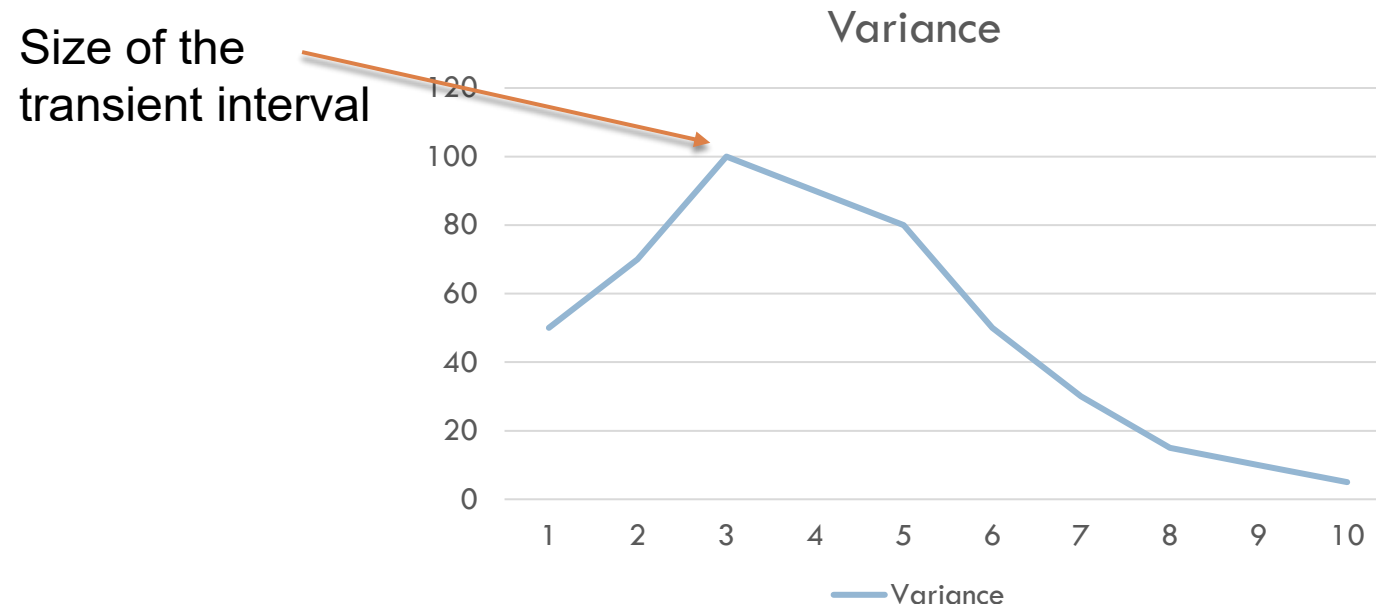
n: amount of observations in a single run.

m: amount of runs (replications).



# Transient elimination

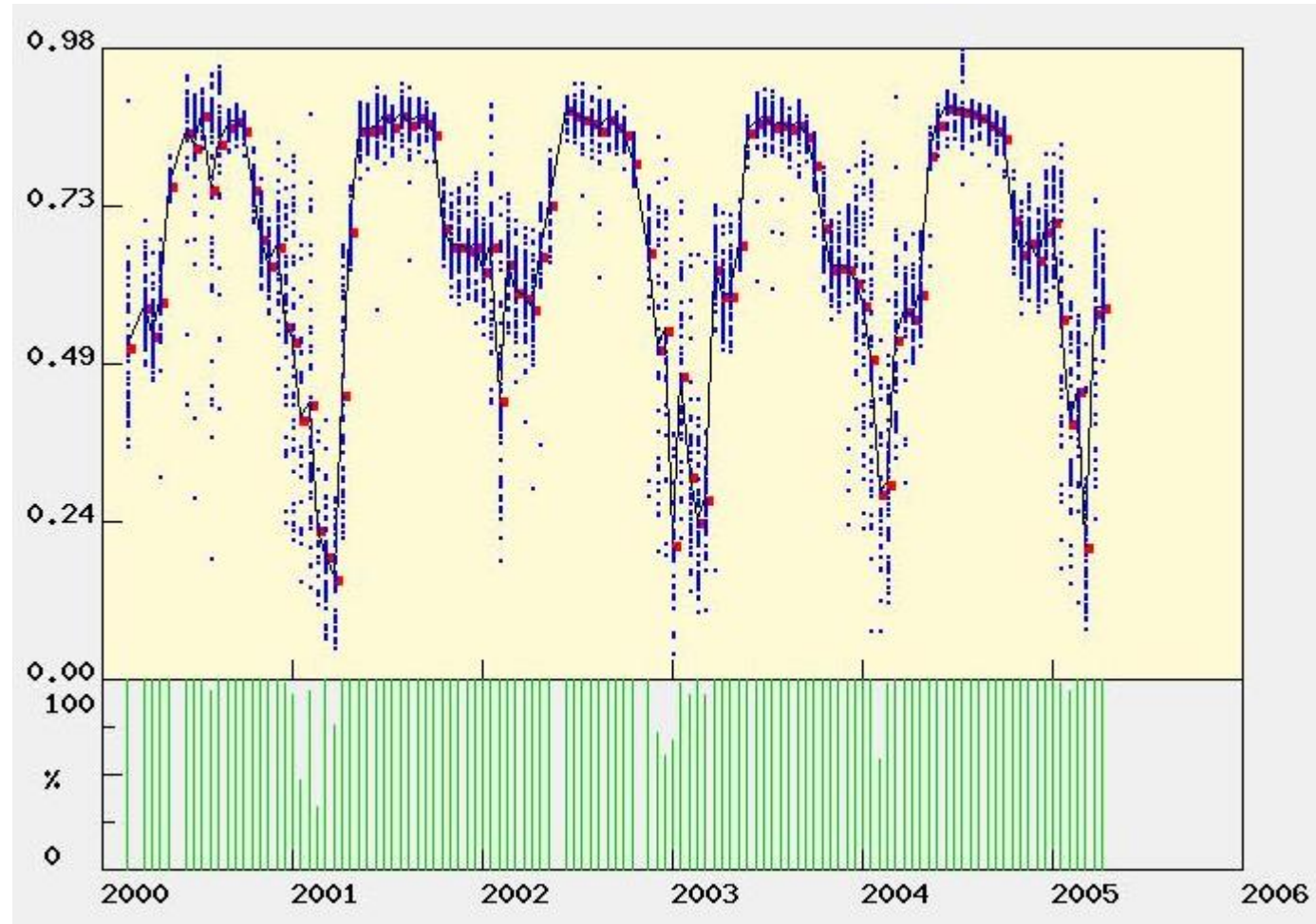
- Step 5: Increase  $n$  by 1 and go to step 2 and plot the variance as a function of  $n$ . The point at which the variance starts to decrease is the length of the transient interval.



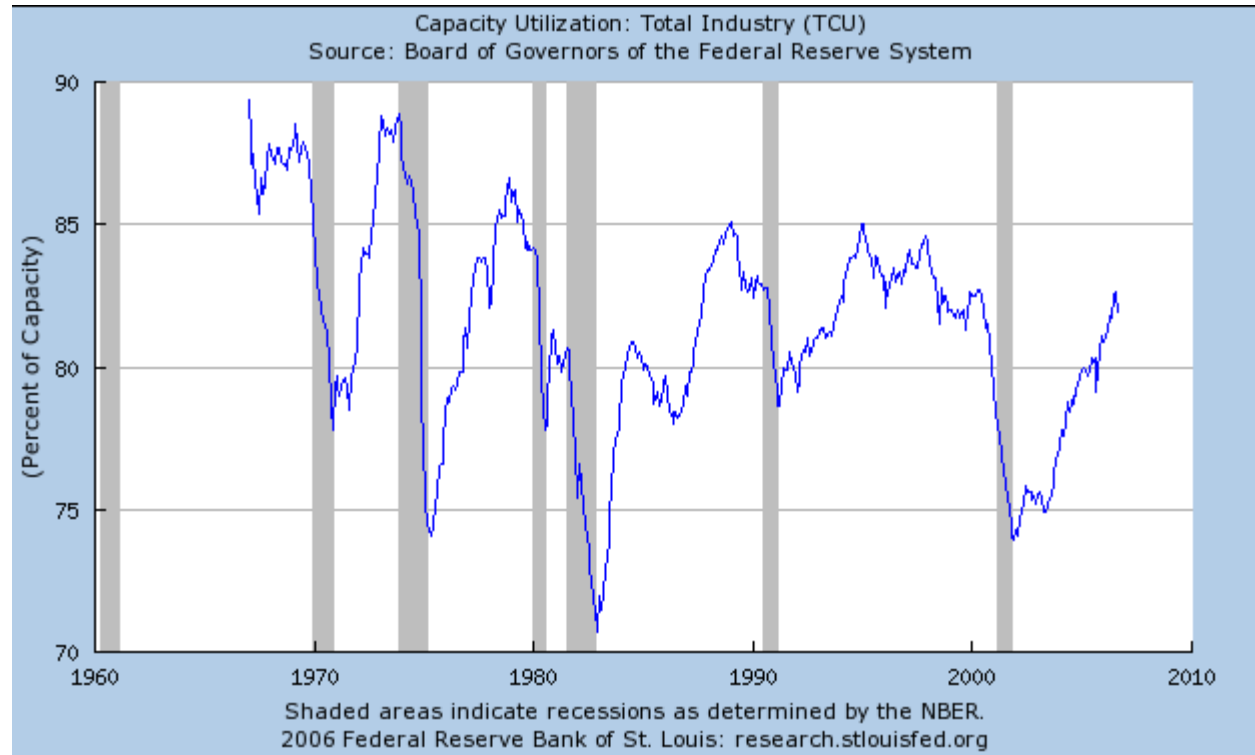
# Regenerative methods

- If the variables observed in the execution of the simulation model, represents, in some way a cyclical restart, that allows suppose the existence of cycles (in the life of the variable). Is likely to consider each one of theses cycles as a replication
- This method is not always applicable. Depends on the existence of cycles in the variables. Also the longitude of this replications must be small; if the longitude of this cycles is big we obtain a small sum of replications.

# Regenerative methods

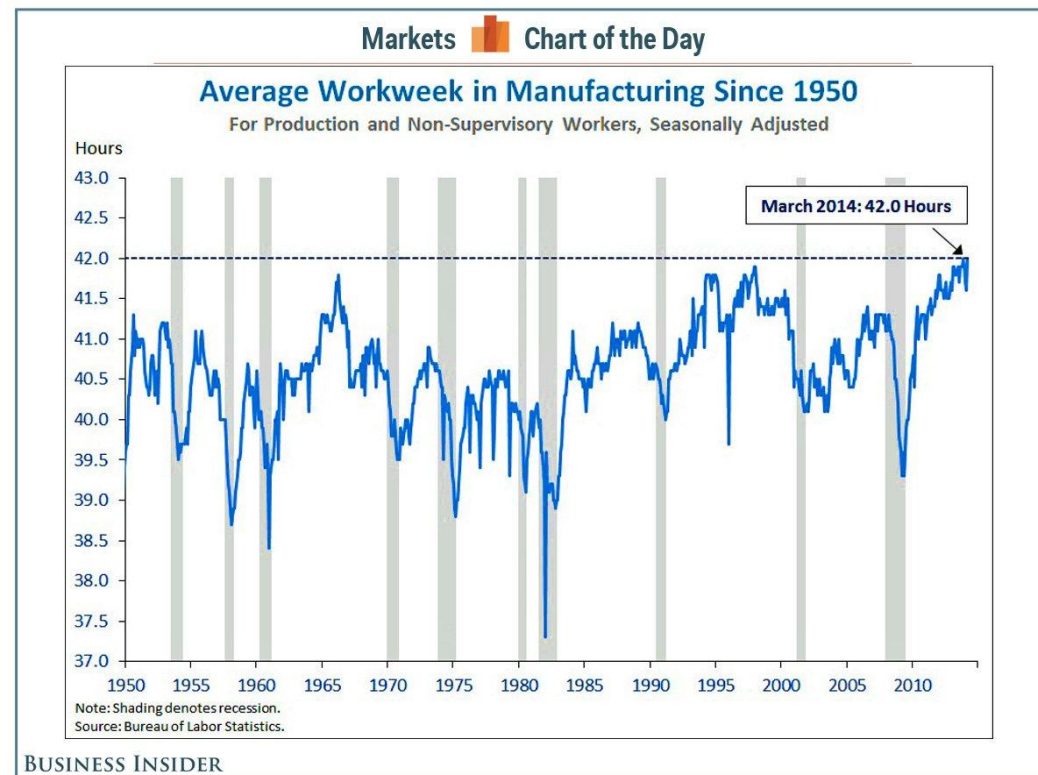


# Regenerative methods



# Regarding the chart

- Nobel laureate and Yale University economist Robert Shiller is in the camp of experts who believe the odds of a recession are very low.
  - ▣ Read more: <http://www.businessinsider.com/shiller-chart-shows-why-recession-is-years-away-2014-4#ixzz30lcrlnl2>



# Regenerative methods

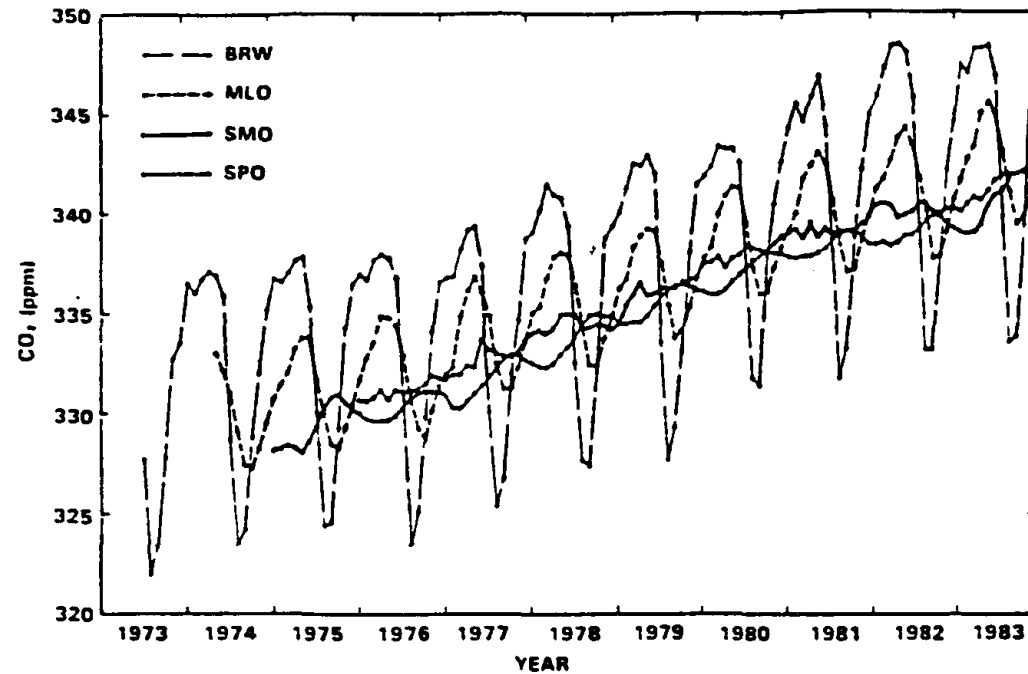


Fig. 1. Selected monthly mean carbon dioxide concentrations from continuous measurements (Barrow, Alaska (BRW); Mauna Loa, Hawaii (MLO); American Samoa (SMO); South Pole (SPO). From: WMO, 1985.

# Applicability

|                         | Terminating simulations                                            | Nonterminating simulations |
|-------------------------|--------------------------------------------------------------------|----------------------------|
| Loading period needed   | Independent repetitions                                            | Independent repetitions    |
| Loading period unneeded | Independent repetitions erasing the loading period/<br>Batch means | Batch means                |



# Variance reduction techniques

Reduce the number of replications



# Motivation

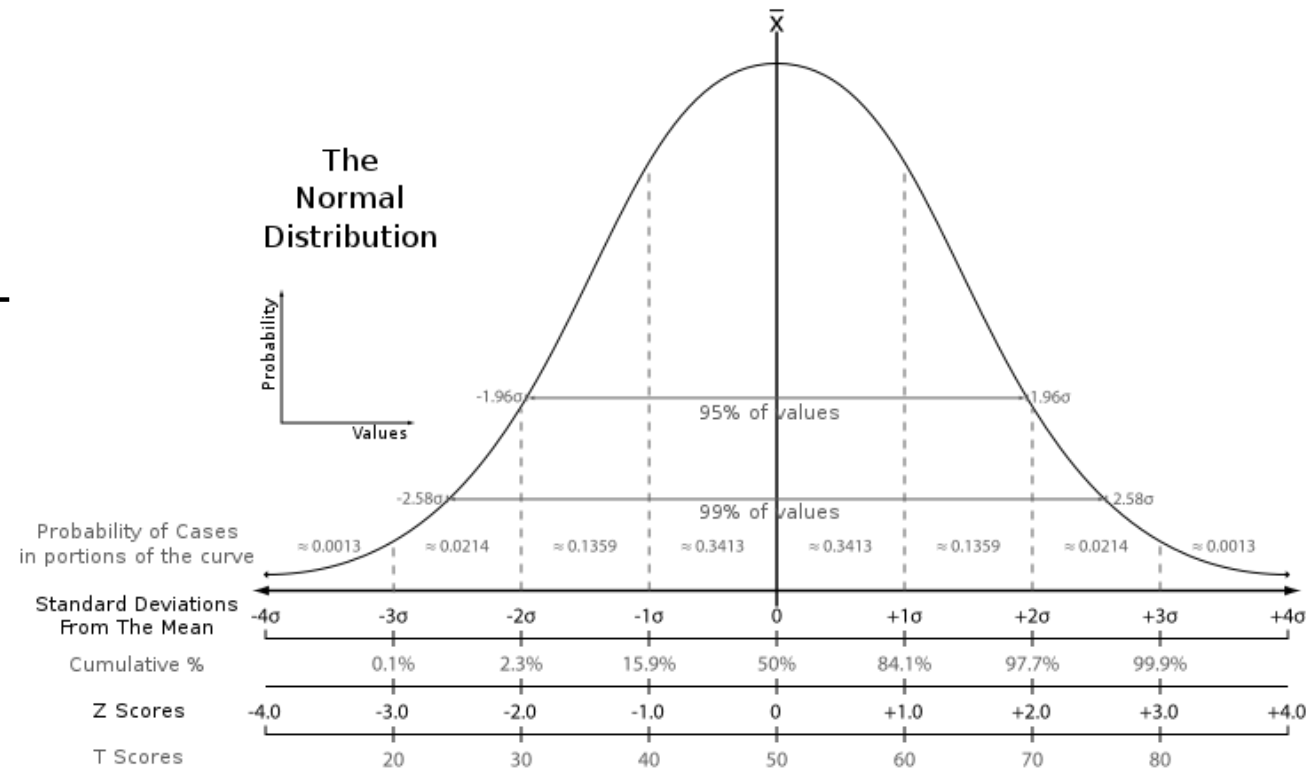
- Interest to reduce the variability introduced in the answer variable due to the use of RNG.
- The value that estimates a specific answer variable, that is represented by its confidence interval, must be adjusted (as possible).

$$(\bar{x} - k \frac{s}{\sqrt{n}}, \bar{x} + k \frac{s}{\sqrt{n}})$$

# Motivation

- Obviously, increasing  $n$ , that is the number of observations, the standard error decreases. Variance reduction techniques try to reduce this variability without the need of increase the number of observations.

$$\frac{s}{\sqrt{n}}$$



# Antithetic variables

- Use of antithetic values o the random numbers stream used.
- In the first execution the random numbers used can be  $(a, b, c, ..) \in [0,1)$ .
- In the second execution we use its antithetic values, that means  $(1-a, 1-b, 1-c, ..) \in [0,1)$ .
- Is needed to establish a synchronization method between both streams

# Control variables

- Simulation allows the observation of the system evolution during the execution of the experiment.
- This allows, in certain grade, to compare the values of the answer variables with the observed values.
- We can add modification to the model to reduce the difference.



# Signification of the effects

The variability is now on the effects

# Have the effects significations?

- The effects have been calculated using the answers that owns variability.
  - ▣ The variability goes to the effects.
    - $\hat{A} = \bar{y}(+) - \bar{y}(-)$

# Have signification?

- From the variance of the effects.
  - ▣ If we have replications, or if we know an estimation of the answer's variance.
- Using the normal distribution (and its probabilistic features).
  - ▣ If we do not have replications and we do not have any estimation of the answer's variance.

# We have replications

## □ Calculation of the variance

- ▣ assuming that the number of replications in each scenario is equal.

- $S_R^2 = \frac{s_1^2 + s_2^2 + \dots + s_N^2}{N}$

- ▣ assuming that the number of replications in each scenario is different.

- $S_R^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2 + \dots + (n_N-1)s_N^2}{n_1 + n_2 + \dots + n_N - N},$

- ▣ Where  $N = 2^k$ , and  $N_t = v2^k$ .



# We have replications

- Variance for an effect

- $\widehat{effect} = \overline{y_+} - \overline{y_-}$

- Where  $\overline{y_+}$  is the mean of the  $N_T/2$  scenarios (+ sign), and  $\overline{y_-}$  is the mean of the  $N_T/2$  scenarios (- sign).

- The calculus is:

- $v(\widehat{effect}) = v(\overline{y_+} - \overline{y_-}) = \frac{S^2}{N_T/2} + \frac{S^2}{N_T/2} = \frac{4S^2}{N_T}$

# Example

| x1 | x2 | Mean         | S2   |  |          |          |
|----|----|--------------|------|--|----------|----------|
| 60 | 64 | 62           | 8    |  |          |          |
| 72 | 74 | 73           | 2    |  |          |          |
| 54 | 55 | 54,5         | 0,5  |  |          |          |
| 68 | 70 | 69           | 2    |  | v(efect) | 12,1942  |
| 52 | 54 | 53           | 2    |  |          | 3,492019 |
| 83 | 87 | 85           | 8    |  |          |          |
| 45 | 50 | 47,5         | 12,5 |  |          |          |
| 80 | 85 | 82,5         | 12,5 |  |          |          |
|    |    | S2= 24,38839 |      |  |          |          |

# Signification test

- Hypothesis testing

- $H_0: \text{effect}=0$

- $H_1: \text{effect} \neq 0$

- Statistic of the test

- $t = \frac{\widehat{effect}-0}{S_{effect}}$

- t-Student with  $n_1+n_2+..+n_n$ -N degrees of freedom ( $N=2^k$ )

- Calculus of the I.C.

- $\widehat{effect} \pm x * S_{effect}$



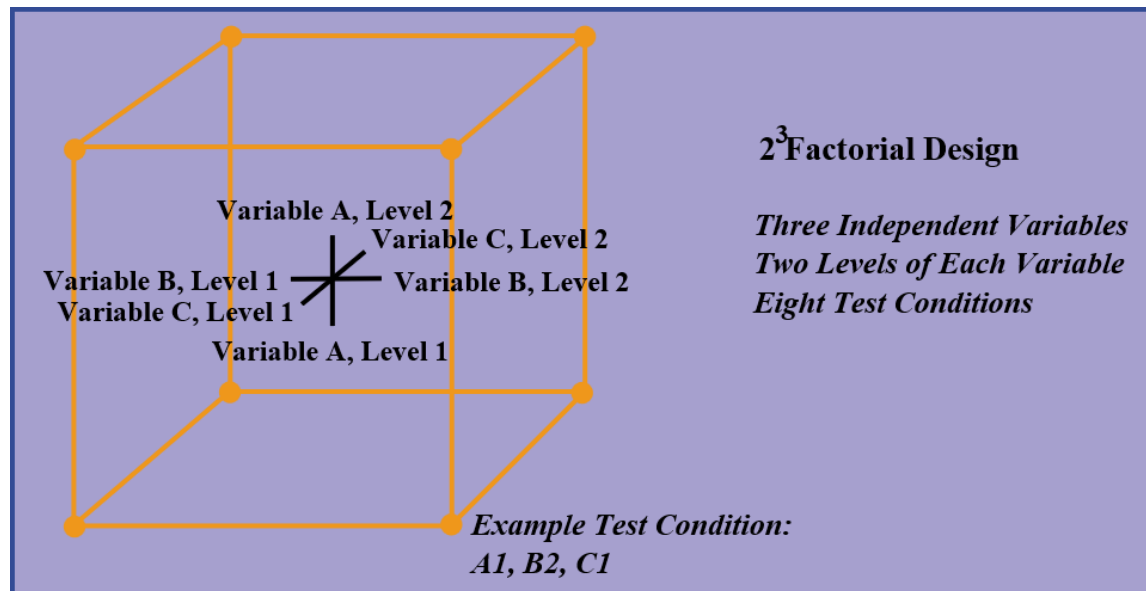
# Fractional factorial design

# Fractional factorial design

- A factorial experiment in which only an **adequately chosen fraction** of the treatment combinations (scenarios) required for the complete factorial experiment **is selected to be run**.
- Hierarchical Ordering principle:
  - ▣ Lower order effects are more likely to be important than higher order effects.
  - ▣ Effects of the same order are equally likely to be important
- Effect Sparsity Principle (Pareto principle)
  - ▣ The number of relatively important effects in a factorial experiment is small
- Effect Heredity Principle
  - ▣ In order for an interaction to be significant, at least one of its parent factors should be significant.

# A $2^{3-1}$ design (half of a $2^3$ )

- Consider the two-level, full factorial design for three factors, namely the  $2^3$  design. This implies eight runs (not counting replications or center points).



# $2^3$ Two-level, Full Factorial Design

|   | X1 | X2 | X3 | Y          |
|---|----|----|----|------------|
| 1 | -1 | -1 | -1 | $y_1 = 33$ |
| 2 | +1 | -1 | -1 | $y_2 = 63$ |
| 3 | -1 | +1 | -1 | $y_3 = 41$ |
| 4 | +1 | +1 | -1 | $y_4 = 57$ |
| 5 | -1 | -1 | +1 | $y_5 = 57$ |
| 6 | +1 | -1 | +1 | $y_6 = 51$ |
| 7 | -1 | +1 | +1 | $y_7 = 59$ |
| 8 | +1 | +1 | +1 | $y_8 = 53$ |

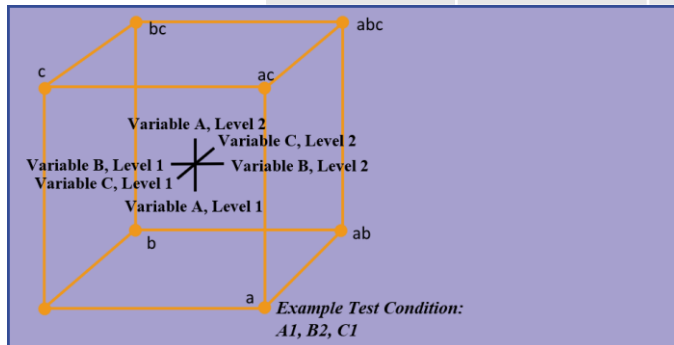
# Computing the effects

- Effect of  $X_1 = (1/4)(y_2 + y_4 + y_6 + y_8) - (1/4)(y_1 + y_3 + y_5 + y_7)$
- $X_1 = (1/4)(63 + 57 + 51 + 53) - (1/4)(33 + 41 + 57 + 59) = 8.5$
- Suppose, however, that we only have enough resources to do four runs. It is still possible to estimate the main effect for  $X_1$ ? Or any other main effect?
  - ▣ The answer is yes, and there are even different choices of the four runs that will accomplish this.



# Only 4 runs

|    |   | C1 | C2 | C3 | Y          |
|----|---|----|----|----|------------|
| Y1 | 1 | -1 | -1 | -1 | $y_1 = 33$ |
| Y2 | 2 | +1 | -1 | -1 | $y_2 = 63$ |
| Y3 | 3 | -1 | +1 | -1 | $y_3 = 41$ |
| Y4 | 4 | +1 | +1 | -1 | $y_4 = 57$ |
| Y5 | 5 | -1 | -1 | +1 | $y_5 = 57$ |
| Y6 | 6 | +1 | -1 | +1 | $y_6 = 51$ |
| Y7 | 7 | -1 | +1 | +1 | $y_7 = 59$ |
| Y8 | 8 | +1 | +1 | +1 | $y_8 = 53$ |



# Main effects

## □ C1 main effect:

$$\blacksquare c1 = (1/2) (y4 + y6) - (1/2) (y1 + y7)$$

$$\blacksquare c1 = (1/2) (57+51) - (1/2) (33+59) = 8$$

## □ C2 main effect

$$\blacksquare c2 = (1/2) (y4 + y7) - (1/2) (y1 + y6)$$

$$\blacksquare c2 = (1/2) (57+59) - (1/2) (33+51) = 16$$

## □ C3 main effect

$$\blacksquare c3 = (1/2) (y6 + y7) - (1/2) (y1 + y4)$$

$$\blacksquare c3 = (1/2) (51+59) - (1/2) (33+57) = 10$$

# Selecting the experiments to execute

- Note that, mathematically,  $2^{3-1} = 2^2$

|   | X1 | X2 |
|---|----|----|
| 1 | -  | -  |
| 2 | +  | -  |
| 3 | -  | +  |
| 4 | +  | +  |

# Adding the column of the interactions

- We add a new column that represents the interactions between X1 and X2

|   | X1 | X2 | X1*X2 |
|---|----|----|-------|
| 1 | -  | -  | +     |
| 2 | +  | -  | -     |
| 3 | -  | +  | -     |
| 4 | +  | +  | +     |

# Adding the column for the new factor

- Now we can substitute this new column for X3

|   | X1 | X2 | X3 |
|---|----|----|----|
| 1 | -  | -  | +  |
| 2 | +  | -  | -  |
| 3 | -  | +  | -  |
| 4 | +  | +  | +  |

# Example

---

- We have 4 factors P, T, D, E.
- $2^4$ .
- We want to perform at maximum 8 experiments.

# Example

|   | P | T | D | $E=P*T$ |
|---|---|---|---|---------|
| 1 | - | - | - | +       |
| 2 | + | - | - | -       |
| 3 | - | + | - | -       |
| 4 | + | + | - | +       |
| 5 | - | - | + | +       |
| 6 | + | - | + | -       |
| 7 | - | + | + | -       |
| 8 | + | + | + | +       |

# Confounding

---

- A confounding design is one where some treatment effects (main or interactions) are estimated by the same linear combination of the experimental observations as some blocking effects.



# The price

- One price we pay for using the design table column  $X1 * X2$  to obtain column  $X3$  is, clearly, our **inability** to obtain an **estimate of the interaction effect** for  $X1 * X2$  (i.e.,  $c12$ ) that is separate from an estimate of the main effect for  $X3$ .
- We have **confounded** the **main effect** estimate for factor  $X3$  (i.e.,  $c3$ ) with the **estimate of the interaction effect** for  $X1$  and  $X2$  (i.e., with  $c12$ )

# Notation

□  $X_3 = X_1 * X_2$  can be represented by:

□  $3=12$

□ Playing with this

□ Multiplying with 3

□  $33=123$ , and  $33=I$  (identity)

□  $I=123$ ,  $2I=2123$ ,  $2I=2$ ,  $I=22$

3

□  $I=123$  is the design generator

□  $1=23$ ,  $2=13$ ,  $3=12$ ,  $I=123$  aliases

# Principal fraction

- We can replace any design generator by its negative counterpart and have an equivalent, but different fractional design.
- The fraction generated by positive design generators is sometimes called the principal fraction.

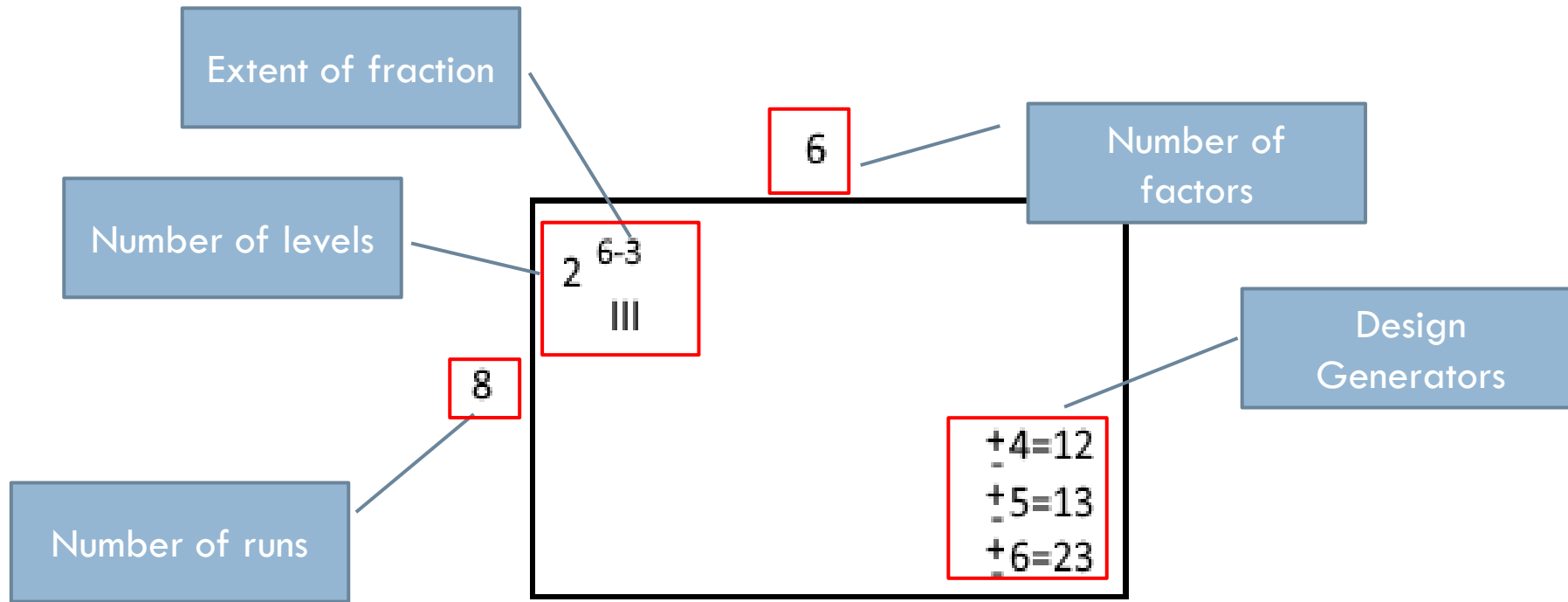
# Confounding pattern

- The confounding pattern described by  $1=23$ ,  $2=13$ , and  $3=12$  tells us that all the main effects of  $2^{3-1}$  design are confounded with two-factor interactions.
  - ▣ That is the price we pay for using this fractional design.

# Confounding pattern

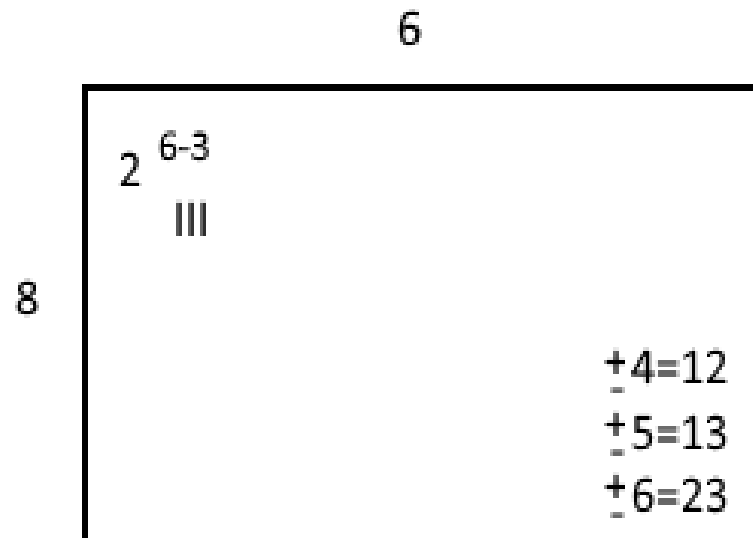
- In the typical quarter-fraction of a  $2^6$  design, i.e., in a  $2^{6-2}$  design, main effects are confounded with three-factor interactions (e.g.,  $5=123$ ) and so on.
- In the case of  $5=123$ , we can also readily see that  $15=23$  (etc.), which alerts us to the fact that certain two-factor interactions of a  $2^{6-2}$  are confounded with other two-factor interactions.

# Definition of the experiment



<https://www.itl.nist.gov/div898/handbook/pri/section3/pri3344.htm>

# How to construct this experiment?

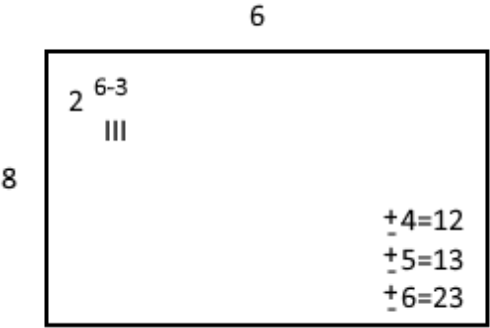


# Construct a Fractional Factorial Design From the Specification

- **Write down a full factorial design** in standard order for  $k-p$  factors ( $6-3 = 3$  factors for the example above). In the specification above we start with a  $2^3$  full factorial design. Such a design has  $2^3 = 8$  rows.
- **Add a sixth column** to the design table for factor 6, using  $4 = 12$  to manufacture it (i.e., create the new column by multiplying the indicated old columns together).
- **Do likewise** for factor 5 and for factor 6, using the appropriate design generators.
- The resultant design matrix gives the **8 trial runs** for an 6-factor fractional factorial design.

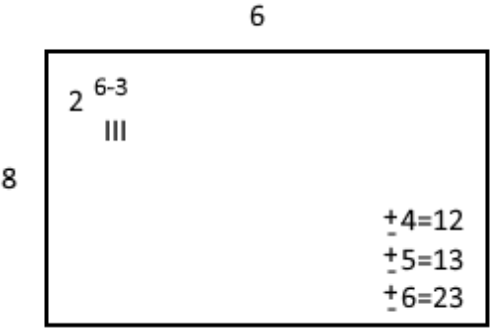


# Example: $2^{6-3}$



| X1 | X2 | X3 | X4 | X5 | X6 |
|----|----|----|----|----|----|
| -1 | -1 | -1 |    |    |    |
| 1  | -1 | -1 |    |    |    |
| -1 | 1  | -1 |    |    |    |
| 1  | 1  | -1 |    |    |    |
| -1 | -1 | 1  |    |    |    |
| 1  | -1 | 1  |    |    |    |
| -1 | 1  | 1  |    |    |    |
| 1  | 1  | 1  |    |    |    |

# Example: $2^{6-3}$



| X1 | X2 | X3 | X4 | X5 | X6 |
|----|----|----|----|----|----|
| -1 | -1 | -1 | 1  | 1  | 1  |
| 1  | -1 | -1 | -1 | -1 | 1  |
| -1 | 1  | -1 | -1 | 1  | -1 |
| 1  | 1  | -1 | 1  | -1 | -1 |
| -1 | -1 | 1  | 1  | -1 | -1 |
| 1  | -1 | 1  | -1 | 1  | -1 |
| -1 | 1  | 1  | -1 | -1 | 1  |
| 1  | 1  | 1  | 1  | 1  | 1  |

# Words

- There are seven "words", or strings of numbers, in the defining relation for the  $2^{8-3}$  design, starting with the original three generators and adding all the new "words" that can be formed by multiplying together any two or three of these original three words. These seven turn out to be  $I = 3456 = 12457 = 12358 = 12367 = 12468 = 3478 = 5678$ .
- In general, there will be  $(2^p - 1)$  words in the defining relation for a  $2^{k-p}$  fractional factorial.

# Resolution

- The **length** of the shortest word in the defining relation is called the resolution of the design.
- Resolution describes the degree to which estimated main effects are confounded (or aliased) with estimated 2-level interactions, 3-level interactions, etc.
- Resolution is added as a Roman numeral to the experiment definition.

# Wordlength

- Alias Structure for  $2_{IV}^{6-2}$  with  $I = ABCE = BCDF = ADEF$
- $I = ABCE = BCDF = ADEF$  implies  $A = BCE = ABCDF = DEF$
- Wordlength pattern  $W = (W_0, W_1, \dots, W_6)$ , where  $W_i$  is the number of defining words of length  $i$  (i.e., involving  $i$  factors)
  - ▣  $W = (1, 0, 0, 0, 3, 0, 0)$
  - ▣ Resolution is the smallest  $i$  such that  $i > 0$  and  $W_i > 0$ . Hence it is a  $2_{IV}^{6-2}$  design

# Wordlength

- $2_{III}^{6-2}$  an alternative design.
- Basic Design: A, B, C, D
- $E = ABCD$ ,  $F = ABC$ , i.e.,  $I = ABCDE$ , and  $I = ABCF$
- which induces:  $I = DEF$  ( $ABCDE=ABCF$ )
- complete defining relation:  $I = ABCDE = ABCF = DEF$
- wordlength pattern:  $W = (1, 0, 0, 1, 1, 1, 0)$

# Minimum Aberration Criterion

- Recall  $2^{k-p}$  with maximum resolution should be preferred. But it is possible that there are two designs that attain the maximum resolution. How should we further distinguish them?
- For example, consider  $2^{7-2}$  fractional factorial design
  - ▣ d1: basic design: A, B, C, D, E; F = ABC, G = ABDE complete defining relation: I = ABCF = ABDEG = CDEFG
    - wordlength pattern:  $W = (1, 0, 0, 0, 1, 2, 0, 0)$
    - Resolution: IV
  - ▣ d2: basic design: A, B, C, D, E; F = ABC, G = ADE complete defining relation: I = ABCF = ADEG = BCDEFG
    - wordlength pattern:  $W = (1, 0, 0, 0, 2, 0, 1, 0)$
    - Resolution: IV

# Minimum Aberration Criterion

- $d1$  and  $d2$ , which is better?
- Minimum Aberration Criterion Definition:
  - ▣ Let  $d1$  and  $d2$  be two  $2^{k-p}$  designs, let  $r$  be the smallest positive integer such that  $W_r(d1) \neq W_r(d2)$ . If  $W_r(d1) < W_r(d2)$ , then  $d1$  is said to have less aberration than  $d2$ .
- If there does not exist any other design that has less aberration than  $d1$ , then  $d1$  has minimum aberration.
- Small Minimum Aberration Designs are used a lot in practice. They are tabulated in most design books. See Table 8-14 in Montgomery. For the most comprehensive table, consult Wu&Hamada.



# Reso

|     | Ability                                                                                                                                                                                                                                                                                | Example                                       |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| I   | Not useful: an experiment of exactly one run only tests one level of a factor and hence can't even distinguish between the high and low levels of that factor                                                                                                                          | $2^1 - 1$ with defining relation $I = A$      |
| II  | Not useful: main effects are confounded with other main effects                                                                                                                                                                                                                        | $2^2 - 1$ with defining relation $I = AB$     |
| III | Estimate main effects, but these may be confounded with two-factor interactions                                                                                                                                                                                                        | $2^3 - 1$ with defining relation $I = ABC$    |
| IV  | Estimate main effects unconfounded by two-factor interactions<br>Estimate two-factor interaction effects, but these may be confounded with other two-factor interactions                                                                                                               | $2^4 - 1$ with defining relation $I = ABCD$   |
| V   | Estimate main effects unconfounded by three-factor (or less) interactions<br>Estimate two-factor interaction effects unconfounded by two-factor interactions<br>Estimate three-factor interaction effects, but these may be confounded with other two-factor interactions              | $2^5 - 1$ with defining relation $I = ABCDE$  |
| VI  | Estimate main effects unconfounded by four-factor (or less) interactions<br>Estimate two-factor interaction effects unconfounded by three-factor (or less) interactions<br>Estimate three-factor interaction effects, but these may be confounded with other three-factor interactions | $2^6 - 1$ with defining relation $I = ABCDEF$ |

# Selection

| Número de factores, k | Fracción  | Número de combinaciones | Generadores de diseño |                  |
|-----------------------|-----------|-------------------------|-----------------------|------------------|
| 3                     | $2^{3-1}$ | 4                       | C=AB                  |                  |
| 4                     | $2^{4-1}$ | 8                       | D=ABC                 |                  |
| 5                     | $2^{5-1}$ | 16                      | E=ABCD                |                  |
|                       | $2^{5-2}$ | 8                       | D=AB                  | E=AC             |
| 6                     | $2^{6-1}$ | 32                      | F=ABCDE               |                  |
|                       | $2^{6-2}$ | 16                      | E=ABC                 | F=BCD            |
|                       | $2^{6-3}$ | 8                       | D=AB<br>E=AC          | F=BC             |
| 7                     | $2^{7-1}$ | 64                      | G=ABCDEF              |                  |
|                       | $2^{7-2}$ | 32                      | F=ABCD                | G=ABDE           |
|                       | $2^{7-3}$ | 16                      | E=ABC<br>F=BCD        | G=ACD            |
|                       | $2^{7-4}$ | 8                       | D=AB<br>E=AC          | F=BC<br>G=ABC    |
| 8                     | $2^{8-2}$ | 64                      | G=ABCD                | H=ABEF           |
|                       | $2^{8-3}$ | 32                      | F=ABC<br>G=ABD        | H=BCDE           |
|                       | $2^{8-4}$ | 16                      | E=BCD<br>F=ACD        | G=ABC<br>H=ABD   |
| 9                     | $2^{9-2}$ | 128                     | H=ACDFG               | J=BCEFG          |
|                       | $2^{9-3}$ | 64                      | G=ABCD<br>H=ACEF      | J=CDEF           |
|                       | $2^{9-4}$ | 32                      | F=BCDE<br>G=ACDE      | H=ABDE<br>J=ABCE |
|                       | $2^{9-5}$ | 16                      | E=ABC                 | H=ABD            |
|                       |           |                         | F=BCD<br>G=ACD        | J=ABCD           |

# Complete definition of the DOE





- We have a limited budget to analyze the different factors to consider on our model. Each individual experiment costs 100€ and we have a total budget of 20.000€ to be destined to experimentation. Define an experiment design, with this constraint, considering that we need at least 3 replications for each experiment.

| GORE                  | C1:<br>BUSH | C2:<br>BUCHANAN | C3:<br>NADER | C4:<br>BROWNE | C5:<br>HAGELIN | C6:<br>HARRIS | C7:<br>MCREYNOLDS | C8:<br>MOOREHEAD | C9:<br>PHILLIPS |
|-----------------------|-------------|-----------------|--------------|---------------|----------------|---------------|-------------------|------------------|-----------------|
| ALACHUA               | 47300       | 34062           | 262          | 3215          | 658            | 42            | 4                 | 658              | 2               |
| BAKER                 | 2392        | 5610            | 73           | 53            | 17             | 3             | 0                 | 0                | 3               |
| BAY                   | 18850       | 38637           | 248          | 828           | 171            | 18            | 5                 | 3                | 37              |
| BRADFORD              | 3072        | 5413            | 65           | 84            | 28             | 2             | 0                 | 0                | 3               |
| BREVARD               | 97318       | 115185          | 570          | 4470          | 643            | 39            | 11                | 11               | 76              |
| More than 100 cases.. |             |                 |              |               |                |               |                   |                  |                 |

# Answer

- We consider that GORE categorical variable are not going to be used on our analysis.
- Assuming that we can spend all the budget in an initial experimentation without considering the possible inconvenient that can appear due to possible high variability of certain variables (that lead us to increase the number of needed replications that is initial considered by 3), the maximum amount of experiment is 64:
  - ▣  $64 \text{ experiment} * 3 \text{ replications} * 100\text{€} = 19200\text{€}$

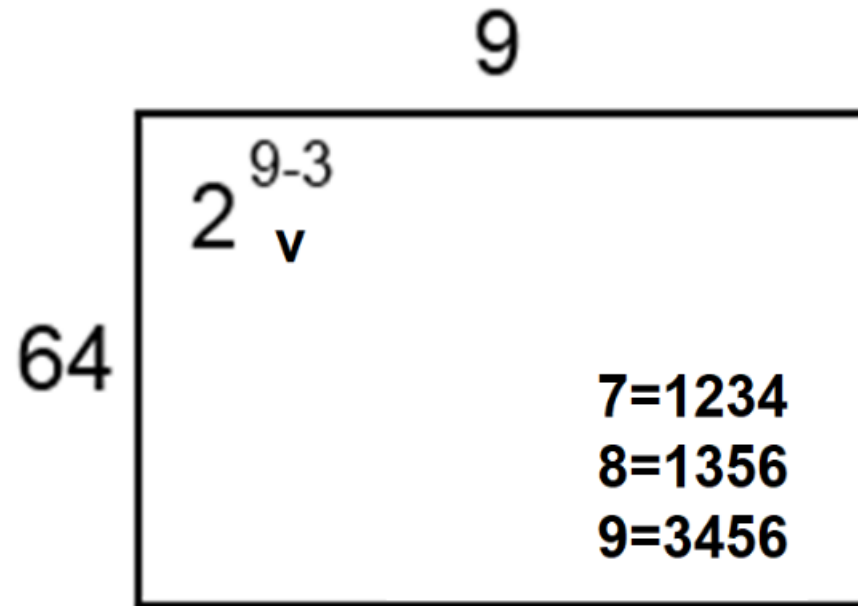
# Answer

- This design don't allow a complete experimentation considering all the factors we have on our model, we have 9 factors that implies a  $2^9$  experiments meaning 512 experiments with 3 replications each one of them.
- This implies that we need to reduce the amount of variables to be considered using a fractional factorial design.

# Answer

- To do this it is needed to define a confounding pattern and select those factors that are going to be confounded.
- The design will be defined as:
  - ▣  $2^{9-3}$
- Hence 3 confounding patterns must be defined.
- We select in our case the last three variables to be confounded using the first 6 variables.

# Answer







# Factorial Design with Foldover

Breaking Aliasing in Fractional Factorial Experiments

# Recap – Fractional Factorial & Aliasing

- Fractional factorial (e.g.,  $2^{k-p}$  )
- Runs only a fraction of all combinations
- Saves time and resources
- The cost: Aliasing (confounding)
  - ▣ Effects become indistinguishable
- Example: In  $2^{3-1}$ ,  $A=BC$ ,  $B=AC$ ,  $C=AB$ 
  - ▣ You cannot tell if a significant effect is due to A or to the BC interaction.

# What is Foldover?

- Foldover = Adding a second fraction that is the “mirror image” of the first
- How?
  - ▣ Take the original fraction
  - ▣ Invert the sign of one or more factors to generate a new set of runs
  - ▣ Combine both fractions for analysis
- Result: Broken aliases → clearer effect estimates

# Types of foldover

| Type                    | What IS inverted?             | Effect                                                                                                                               |
|-------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full foldover</b>    | All factors                   | All main effects become unaliased from 2-factor interactions → Resolution increases by 1 (e.g., III → IV, or even to full factorial) |
| <b>Partial foldover</b> | A single factor (or a subset) | Only that factor's main effect is separated from its 2-fi aliases                                                                    |

# Example – $2^{3-1}$ Design (4 runs)

- First fraction (generator  $I=ABC$ ):

| Run | A | B | C | Response |
|-----|---|---|---|----------|
| 1   | + | + | + | y1       |
| 2   | + | – | – | y2       |
| 3   | – | + | – | y3       |
| 4   | – | – | + | y4       |

- Aliasing:  $A=BC$ ,  $B=AC$ ,  $C=AB$

# Apply Full Foldover (invert ALL factors)

- New fraction (signs flipped):

| Run | A | B | C | Response |
|-----|---|---|---|----------|
| 5   | − | − | − | y5       |
| 6   | − | + | + | y6       |
| 7   | + | − | + | y7       |
| 8   | + | + | − | y8       |

- Combine runs 1-8 → You obtain all 8 combinations of a full  $2^3$  factorial.
- No aliasing remains. Main effects and interactions are estimable separately.

# Key Insight - Cost vs. Benefit

- Original  $2^{3-1}$ : 4 runs (confounded effects)
- After full foldover: 8 runs = full factorial
- In this specific case, foldover reproduces the complete design with exactly the same number of runs as a full factorial.
- In larger designs (e.g.,  $2^{4-1}$ ), foldover gives a resolution IV design with 16 runs instead of 32 (full factorial), still breaking main effect-interaction aliases.

# When to Use Foldover

- You performed a screening design (low resolution) and saw significant effects
- You want to resolve ambiguities without discarding the first fraction
- You have budget for approximately double the runs of the initial fraction
- You need clearer estimates for final optimization or response surface modeling.



# Summary

- Foldover is an efficient, sequential strategy to upgrade design resolution without starting over.

| Concept                  | Takeaway                                                 |
|--------------------------|----------------------------------------------------------|
| <b>Problem</b>           | Fractional factorial → aliased effects                   |
| <b>Foldover</b>          | Run a mirrored second fraction                           |
| <b>Full foldover</b>     | Invert all factors → breaks all main effect-2fi aliases  |
| <b>Partial foldover</b>  | Invert one factor → resolves only that factor's aliasing |
| <b>Example</b> $2^{3-1}$ | Full foldover = full factorial (8 runs)                  |

# If You Only Want Main Effects - Foldover Is NOT Needed, but...

- Key point: In a Resolution III fractional factorial design (e.g.,  $2^{3-1}$ ), main effects are aliased with two-factor interactions.
  - ▣ Example:  $A=BC$ ,  $B=AC$ ,  $C=AB$
- But, if you can safely assume that all two-factor interactions are negligible (zero), then:
  - ▣ The alias  $A=BC$  means estimated effect = true A + true BC
  - ▣ If  $BC \approx 0$ , then the estimate  $\approx$  true A
  - ▣ You can interpret the estimated effects as pure main effects without running a second fraction
- Therefore: No foldover  $\rightarrow$  only the initial 4 runs for  $2^{3-1}$
- Foldover  $\rightarrow$  8 runs (full factorial)

# If You Only Want Main Effects - Foldover Is NOT Needed, but...

- When do you need foldover?. When you suspect interactions might be significant, or you want to prove they are negligible rather than assume it.
- Takeaway: Foldover is for resolving ambiguity, not for estimating main effects alone if you are willing to assume no interactions.

| Scenario                                        | Recommended action                           |
|-------------------------------------------------|----------------------------------------------|
| Main effects only, interactions believed absent | Fractional factorial <b>without foldover</b> |
| Main effects + possible interactions            | Foldover (or full factorial)                 |



# Blocking in Factorial Designs

Controlling Nuisance Variables

# Blocking vs. Foldover - Different Objectives

- **Key takeaway:** Blocking **hides** nuisance variation. Foldover **reveals** hidden interactions.

| Aspect                     | Blocking                                                                      | Foldover                                                             |
|----------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Primary goal</b>        | Remove variability from a known nuisance factor                               | Break unwanted aliasing between main effects and interactions        |
| <b>When to use</b>         | When you have a factor, you don't care about but affects response             | When you have a fractional factorial and want to resolve ambiguities |
| <b>What is confounded?</b> | Interactions (or main effects) are <b>deliberately</b> confounded with blocks | Aliases are <b>removed</b> by adding a mirrored fraction             |
| <b>Typical design</b>      | Full or fractional factorial with blocking                                    | Fractional factorial + second fraction                               |
| <b>Result</b>              | More precise estimates of factor effects                                      | Clear separation of main effects from 2-factor interactions          |

# What is Blocking?

- **Blocking** = Grouping experimental runs into homogeneous sets (blocks) to account for a nuisance variable.
  - ▣ The nuisance variable is not of interest but may influence the response.
  - ▣ By **confounding** (aliasing) some effects with blocks, we remove block-to-block variability from the error term.

# Scenario: Testing a New Assembly Line Tool

- Imagine you are a plant manager at an automotive factory. You want to test if a new, ergonomically designed wrench (**Tool B**) allows workers to assemble car doors faster than the current standard wrench (**Tool A**).
- However, you know that **human productivity fluctuates depending on the day of the week**:
  - ▣ **Mondays** are notoriously slow because workers are getting back into the groove after the weekend.
  - ▣ **Fridays** can be erratic because people are tired or rushed to finish the week.
  - ▣ **Wednesdays** are usually steady.
- If you test Tool A on Monday and Tool B on Wednesday, Tool B will look much better simply because of the day it was tested, not because it's actually a better tool.

# Scenario: Testing a New Assembly Line Tool

- **The Blocking Solution**
- To prevent the "day-of-the-week effect" from ruining your data, you use **Day** as your **blocking factor**.
  - ▣ **The Blocks:** Monday, Tuesday, Wednesday (each day is a block).
  - ▣ **The Treatment:** Tool A (Standard) vs. Tool B (New).
- Within each single day, you randomly assign which shifts or workers use Tool A and which use Tool B. This ensures both tools face the exact same "Monday fatigue" or "Wednesday peak productivity."



# Randomized Complete Block Design (RCBD)

| Block (Day of the Week)   | Experimental Units (Shifts) | Treatment 1 (Tool A - Standard) | Treatment 2 (Tool B - New)   |
|---------------------------|-----------------------------|---------------------------------|------------------------------|
| <b>Block 1: Monday</b>    | Shift 1 & Shift 2           | Assigned randomly to a shift    | Assigned randomly to a shift |
| <b>Block 2: Tuesday</b>   | Shift 3 & Shift 4           | Assigned randomly to a shift    | Assigned randomly to a shift |
| <b>Block 3: Wednesday</b> | Shift 5 & Shift 6           | Assigned randomly to a shift    | Assigned randomly to a shift |

# Balanced Incomplete Block Design (BIBD)

- When you have a limitation on how many runs you can perform at one time (due to time, shift lengths, or equipment availability), you *must* use blocking. If you don't, the differences between Day 1 and Day 2 will mix up with your results.
- Depending on how many treatments you are testing, this scenario usually falls into one of two designs:
- **Scenario A: You have 2 Treatments (The Perfect Fit)**
- If you are testing **2 treatments (A and B)** and you want to replicate each one 4 times (making 8 runs total), you can fit a **Complete Block** into each day.
  - ▣ **Day 1 (Block 1):** You run 2 tests of Treatment A and 2 tests of Treatment B (in random order).
  - ▣ **Day 2 (Block 2):** You run 2 tests of Treatment A and 2 tests of Treatment B (in random order).
- Because both treatments are fully represented on both days, you can perfectly subtract the "day effect" from your analysis.

# Balanced Incomplete Block Design (BIBD)

- **Scenario B: You have 8 *Different* Treatments (Balanced Incomplete Blocks)**
- If you have **8 entirely different treatments** (e.g., 8 different chemical formulas) and you can only do 4 per day, you cannot fit them all into one day. This is called an **Incomplete Block Design**.

# Balanced Incomplete Block Design (BIBD)

- A **Balanced Incomplete Block Design (BIBD)** is used when you cannot test all treatments within a single block due to resource, space, or time constraints, but you still want to maintain a perfectly fair and balanced comparison between all treatments.
- To be a BIBD, the design must satisfy two conditions:
  - ▣ Every treatment is repeated the exact same number of times ( $r$ ).
  - ▣ Every possible pair of treatments appears together in a block the exact same number of times ( $\lambda$ ).

# Balanced Incomplete Block Design (BIBD)

- **A Concrete Example: Testing 4 Web Page Designs**
- Imagine you are a UX researcher testing **4 different landing page designs** (Treatments: **A, B, C, D**) to see which one gets the highest conversion rate.
- However, a user test session is short, and you determine that a single participant (Block) only has the patience to evaluate **2 designs** without getting fatigued.
  - ▣ **Total Treatments (v):** 4 (A, B, C, D)
  - ▣ **Block Size (k):** 2 (Max designs per person)
  - ▣ **Total Blocks (b):** 6 participants are needed to make the math perfectly balanced.
- Using the BIBD formula, this setup results in:
  - ▣ Each design being tested **3 times** ( $r=3$ ).
  - ▣ Each pair of designs (e.g., A vs B, A vs C) appearing together exactly **1 time** ( $\lambda=1$ ).

# Balanced Incomplete Block Design (BIBD)

- Here is how you would layout the experiment across your 6 participants (blocks). The order in which each participant sees their two assigned pages must be randomized.

| Block (Participant) | Design Slot 1 | Design Slot 2 | Pairs Present in this Block |
|---------------------|---------------|---------------|-----------------------------|
| Participant 1       | A             | B             | (A, B)                      |
| Participant 2       | A             | C             | (A, C)                      |
| Participant 3       | A             | D             | (A, D)                      |
| Participant 4       | B             | C             | (B, C)                      |
| Participant 5       | B             | D             | (B, D)                      |
| Participant 6       | C             | D             | (C, D)                      |

# Balanced Incomplete Block Design (BIBD)

- **Why this Incomplete Design is "Balanced"**
- If you count the occurrences in the table, you will notice the elegant mathematical symmetry that defines a BIBD:
  - ▣ Count of **A**: 3 times (Blocks 1, 2, 3)
  - ▣ Count of **B**: 3 times (Blocks 1, 4, 5)
  - ▣ Count of **C**: 3 times (Blocks 2, 4, 6)
  - ▣ Count of **D**: 3 times (Blocks 3, 5, 6)
- More importantly, look at the head-to-head pairs. **How many times does A go up against B?** Just once (Block 1). **How many times does C go up against D?** Just once (Block 6). Every single combination of pairs occurs exactly  $\lambda=1$  time.
- Because of this perfect balance, your statistical software can easily separate the "participant bias" from the "design performance," giving you an incredibly accurate picture of which landing page is truly the best, even though no single person ever saw all four.

# How Blocking Works - Basic Idea

- Without blocking:
  - ▣ All variation between days goes into experimental error → less power.
- With blocking:
  - ▣ Assign runs to days so that **block effects** are separated from treatment effects.
  - ▣ Usually, **interactions of high order** (e.g., ABC) are confounded with blocks because they are unlikely to be significant.



# How to Choose What to Confound with Blocks

- Principle: Confound interactions that are unlikely to be significant (usually 3-way and higher) or that you don't care about.
  - ▣ For screening: main effects are most important, so confound interactions of order  $\geq 2$ .
  - ▣ For optimization: you may want 2-fi clear, so use larger block sizes or foldover within blocks.
- Rule of thumb:
  - ▣ Fewer blocks (larger block size)  $\rightarrow$  confound higher-order interactions.
  - ▣ More blocks (smaller block size)  $\rightarrow$  you may be forced to confound some 2-factor interactions (see table, e.g.,  $k=4$ , 8 blocks  $\rightarrow$  many 2-fi confounded).

# Blocking Table

| k   | # blocks | Block size | Generating effects | Interactions confounded with blocks |
|-----|----------|------------|--------------------|-------------------------------------|
| 3   | 2        | 4          | ABC                | ABC                                 |
| 3   | 4        | 2          | AB, AC             | AB, AC, BC                          |
| 4   | 2        | 8          | ABCD               | ABCD                                |
| 4   | 4        | 4          | ABC, ACD           | ABC, ACD, BD                        |
| 4   | 8        | 2          | AB, BC, CD         | AB, BC, CD, AC, BD, AD, ABCD        |
| ... | ...      | ...        | ...                | ...                                 |

- **Interpretation:** If you need 4 blocks ( $k=3$ ), use AB and AC to generate blocks. Then AB, AC, and BC are confounded with blocks. Main effects A, B, C remain clean.

# Blocking Table

| Número de factores, k | Número de bloques, 2 <sup>p</sup> | Tamaño del bloque, 2 <sup>k-p</sup> | Efectos elegidos para generar los bloques | Interacciones confundidas con los bloques                                                                                         |
|-----------------------|-----------------------------------|-------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 3                     | 2                                 | 4                                   | ABC                                       | ABC                                                                                                                               |
|                       | 4                                 | 2                                   | AB,AC                                     | AB,AC,BC                                                                                                                          |
| 4                     | 2                                 | 8                                   | ABCD                                      | ABCD                                                                                                                              |
|                       | 4                                 | 4                                   | ABC,ACD                                   | ABC,ACD,BD                                                                                                                        |
|                       | 8                                 | 2                                   | AB,BC,CD                                  | AB,BC,CD,AC,BD,AD,ABCD                                                                                                            |
| 5                     | 2                                 | 16                                  | ABCDE                                     | ABCDE                                                                                                                             |
|                       | 4                                 | 8                                   | ABC,CDE                                   | ABC,CDE,ABDE                                                                                                                      |
|                       | 8                                 | 4                                   | ABE,BCE,CDE                               | ABE,BCE,CDE,AC,ABCD,BD,ADE                                                                                                        |
|                       | 16                                | 2                                   | AB,AC,CD,DE                               | Todas las interacciones de dos y cuatros factores                                                                                 |
| 6                     | 2                                 | 32                                  | ABCDEF                                    | ABCDEF                                                                                                                            |
|                       | 4                                 | 16                                  | ABCF,CDEF                                 | ABCF,CDEF,ABDE                                                                                                                    |
|                       | 8                                 | 8                                   | ABEF,ABCD,ACE                             | ABEF,ABCD,ACE,BCF,BDE,CDEF,ADF                                                                                                    |
|                       | 16                                | 4                                   | ABF,ACF,BDF,DEF                           | ABF,ACF,BDF,DEF,BC,ABCD,ABDE;AD;ACDE,CE,BDF,BCDEF,ABCEF,AEF,BE                                                                    |
|                       | 32                                | 2                                   | AB,BC,CD,DE,EF                            | Todas las interacciones de dos, cuatro y seis factores.                                                                           |
| 7                     | 2                                 | 64                                  | ABCDEFG                                   | ABCDEFG                                                                                                                           |
|                       | 4                                 | 32                                  | ABCFG,CDEFG                               | ABCFG,CDEFG,ABDE                                                                                                                  |
|                       | 8                                 | 16                                  | ABC,DEF,AFG                               | ABC,DEF,AFG,ABCDEF,BCFG,ADEG,BCDEG                                                                                                |
|                       | 16                                | 8                                   | ABCD,EFG,CDE,ADG                          | ABCD,EFG,CDE,ADG,ABCDEF;ABE,BCG,CDFG,ADEF,ACEG,ABFG,BCEF,BDEG,ACF,BDF                                                             |
|                       | 32                                | 4                                   | ABG,BCG,CDG,DEG,EFG                       | ABG,BCG,CDG,DEG,EFG,AC,BD,CE,DF,AE,BE,ABCD,ABDE,ABEF,BCDE,BCEF,CDEF,ABCDEF,ADG,ACDEG,ACEFG,ABDFG,ABCEG,BEG,CFG,ADEF,ACDF,ABCF,AFG |
|                       | 64                                | 2                                   | AB,BC,CD,DE,EF,FG                         | Todas las interacciones de dos, cuatro y seis factores.                                                                           |



# Plackett-Burman designs

# Plackett-Burman designs

- In 1946, R.L. Plackett and J.P. Burman published their now famous paper "The Design of Optimal Multifactorial Experiments" in *Biometrika* (vol. 33). This paper described the construction of very economical designs with the run number a multiple of four (rather than a power of 2).
- Plackett-Burman designs are very efficient screening designs when only main effects are of interest.

# Plackett and Burman designs (1946)

- Effects of main factors only
  - ▣ Logically minimal number of experiments to estimate effects of  $m$  input parameters (factors)
  - ▣ Ignores interactions
- Requires  $O(m)$  scenarios
  - ▣ Instead of  $O(2^m)$  or  $O(v^m)$

# Plackett and Burman Designs

- PB designs exist only in sizes that are multiples of 4
- Requires  $X$  experiments for  $m$  parameters
  - ▣  $X = \text{next multiple of } 4 \geq m$
- PB design matrix
  - ▣ Rows = configurations
  - ▣ Columns = factor's values in each configuration
    - High/low =  $+1 / -1$
  - ▣ First row = from P&B paper
  - ▣ Subsequent rows = circular right shift of preceding row
  - ▣ Last row = all  $(-1)$

# Plackett and Burman Designs

- PB designs also exist for 20-run, 24-run, and 28-run (and higher) designs.
- With a 20-run design you can run a screening experiment for up to 19 factors, up to 23 factors in a 24-run design, and up to 27 factors in a 28-run design.



# PB Design Matrix

| Config | Input Parameters (factors) |    |    |    |    |    |    | Response |
|--------|----------------------------|----|----|----|----|----|----|----------|
|        | A                          | B  | C  | D  | E  | F  | G  |          |
| 1      | +1                         | +1 | +1 | -1 | +1 | -1 | -1 |          |
| 2      | -1                         | +1 | +1 | +1 | -1 | +1 | -1 |          |
| 3      | -1                         | -1 | +1 | +1 | +1 | -1 | +1 |          |
| 4      | +1                         | -1 | -1 | +1 | +1 | +1 | -1 |          |
| 5      | -1                         | +1 | -1 | -1 | +1 | +1 | +1 |          |
| 6      | +1                         | -1 | +1 | -1 | -1 | +1 | +1 |          |
| 7      | +1                         | +1 | -1 | +1 | -1 | -1 | +1 |          |
| 8      | -1                         | -1 | -1 | -1 | -1 | -1 | -1 |          |
| Effect |                            |    |    |    |    |    |    |          |

# PB Design Matrix

| Config | Input Parameters (factors) |    |    |    |    |    |    | Response |
|--------|----------------------------|----|----|----|----|----|----|----------|
|        | A                          | B  | C  | D  | E  | F  | G  |          |
| 1      | +1                         | +1 | +1 | -1 | +1 | -1 | -1 | 10       |
| 2      | -1                         | +1 | +1 | +1 | -1 | +1 | -1 | 12       |
| 3      | -1                         | -1 | +1 | +1 | +1 | -1 | +1 | 3        |
| 4      | +1                         | -1 | -1 | +1 | +1 | +1 | -1 | 5        |
| 5      | -1                         | +1 | -1 | -1 | +1 | +1 | +1 | 6        |
| 6      | +1                         | -1 | +1 | -1 | -1 | +1 | +1 | 5        |
| 7      | +1                         | +1 | -1 | +1 | -1 | -1 | +1 | 8        |
| 8      | -1                         | -1 | -1 | -1 | -1 | -1 | -1 | 9        |
| Effect |                            |    |    |    |    |    |    |          |

# PB Design Matrix

| Config | Input Parameters (factors) |    |    |    |    |    |    | Response |
|--------|----------------------------|----|----|----|----|----|----|----------|
|        | A                          | B  | C  | D  | E  | F  | G  |          |
| 1      | +1                         | +1 | +1 | -1 | +1 | -1 | -1 | 10       |
| 2      | -1                         | +1 | +1 | +1 | -1 | +1 | -1 | 12       |
| 3      | -1                         | -1 | +1 | +1 | +1 | -1 | +1 | 3        |
| 4      | +1                         | -1 | -1 | +1 | +1 | +1 | -1 | 5        |
| 5      | -1                         | +1 | -1 | -1 | +1 | +1 | +1 | 6        |
| 6      | +1                         | -1 | +1 | -1 | -1 | +1 | +1 | 5        |
| 7      | +1                         | +1 | -1 | +1 | -1 | -1 | +1 | 8        |
| 8      | -1                         | -1 | -1 | -1 | -1 | -1 | -1 | 9        |
| Effect | -0,5                       |    |    |    |    |    |    |          |

$$-0.5 = (+1(10) + 1(5) + 1(5) + 1(8))/4 - (1(12) + 1(3) + 1(6) + 1(6))/4$$

# PB Design Matrix

| Config | Input Parameters (factors) |     |     |      |      |      |      | Response |
|--------|----------------------------|-----|-----|------|------|------|------|----------|
|        | A                          | B   | C   | D    | E    | F    | G    |          |
| 1      | +1                         | +1  | +1  | -1   | +1   | -1   | -1   | 10       |
| 2      | -1                         | +1  | +1  | +1   | -1   | +1   | -1   | 12       |
| 3      | -1                         | -1  | +1  | +1   | +1   | -1   | +1   | 3        |
| 4      | +1                         | -1  | -1  | +1   | +1   | +1   | -1   | 5        |
| 5      | -1                         | +1  | -1  | -1   | +1   | +1   | +1   | 6        |
| 6      | +1                         | -1  | +1  | -1   | -1   | +1   | +1   | 5        |
| 7      | +1                         | +1  | -1  | +1   | -1   | -1   | +1   | 8        |
| 8      | -1                         | -1  | -1  | -1   | -1   | -1   | -1   | 9        |
| Effect | -0,5                       | 3,5 | 0,5 | -0,5 | -2,5 | -0,5 | -3,5 |          |

# PB Design

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- Magnitude of effect is important, sign is meaningless.
- In the previous example (from most important to least important effects): B, G, E, A, C, D and F.

| Scenarios | Factors | Experiment |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-----------|---------|------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4         | 3       |            | + | + | - |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 8         | 7       |            | + | + | + | - | + | - | - |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 12        | 11      |            | + | + | - | + | + | + | - | - | - | + | - |   |   |   |   |   |   |   |   |   |   |
| 16        | 15      |            | + | + | + | + | - | + | - | + | + | - | - | + | - | - | - |   |   |   |   |   |   |
| 20        | 19      |            | + | + | - | - | + | + | + | + | - | + | - | + | - | - | - | + | + | - |   |   |   |
| 24        | 23      |            | + | + | + | + | + | - | + | - | + | + | - | + | + | - | - | + | - | + | - | - | - |

Plackett RL and Burman J . P. (1946). THE DESIGN OF OPTIMUM MULTIFACTORIAL EXPERIMENTS Design of optimum multifactorial experiments Suppose also that the extreme value of  $x_i$  under consideration is  $x_i$ . Then the main effect of. *Biometrika*, 33(4), 305–325.

# Example

- A statistical approach to the experimental design of the sulfuric acid leaching of gold-copper ore.
- [http://www.scielo.br/scielo.php?script=sci\\_arttext&pid=S0104-66322003000300010](http://www.scielo.br/scielo.php?script=sci_arttext&pid=S0104-66322003000300010)

# Our factors



**Table 4: Mineralogical analysis of transition ore sample and sulfuric acid leach residue**

| Minerals          | Molecular formulae                                                                 | Assay (%)      |               |
|-------------------|------------------------------------------------------------------------------------|----------------|---------------|
|                   |                                                                                    | Transition ore | Leach residue |
| Native Cu         | Cu                                                                                 | 0.39           | 0.30          |
| Chalcopyrite      | CuFeS <sub>2</sub>                                                                 | r              | r             |
| Bornite           | Cu <sub>5</sub> FeS <sub>4</sub>                                                   | r              | r             |
| Chalcocite        | Cu <sub>2</sub> S                                                                  | rr             | rr            |
| Covelite          | CuS                                                                                | rr             | rr            |
| Cuprite           | Cu <sub>2</sub> O                                                                  | t              | -             |
| Malachite         | Cu <sub>2</sub> (CO <sub>3</sub> )(OH) <sub>2</sub>                                | t              | -             |
| Goethite/Limonite | HFeO <sub>2</sub> /Fe <sub>2</sub> O <sub>3</sub> .H <sub>2</sub> O                | 26             | 21            |
| Iron oxide        | Fe <sub>2</sub> O <sub>3</sub> , Fe <sub>3</sub> O <sub>4</sub>                    | 8              | 7             |
| Clorite           | (Mg,Al,Fe) <sub>12</sub> [(Si,Al) <sub>8</sub> O <sub>20</sub> ](OH) <sub>16</sub> | 33             | 35            |
| Quartz            | SiO <sub>2</sub>                                                                   | 26             | 33            |

Notations -: not detected rr: very rare (some cristals) r: rare (~0.2%) t: trace (~0.5%) <1: ~0.8%





# Latin squares

More than two levels.



Fisher window  
Anders Sandberg

# When to apply?

- Latin square designs are used when the factors of interest have more than two levels ( $k$ ).
- We have tree factors.
- There are no (or only negligible) interactions between factors.

# Pros and cons

## □ PROS:

- ▣ They handle the case when we have several nuisance factors and we either cannot combine them into a single factor or we wish to keep them separate.
- ▣ They allow experiments with a relatively small number of runs.

## □ CONS:

- ▣ The number of levels of each blocking variable must equal the number of levels of the treatment factor.
- ▣ The Latin square model assumes that there are no interactions between the blocking variables or between the treatment variable and the blocking variable.

# Example

- What is the best (fast) browser depending on the OS and the computer?
- Browsers are labeled by A,B, C and D.

| Computer | OS |   |   |   |
|----------|----|---|---|---|
|          | 1  | 2 | 3 | 4 |
| I3       | A  | B | C | D |
| I5       | B  | C | D | A |
| I7       | C  | D | A | B |
| XEON     | D  | A | B | C |

# Terminology

- A Latin Square is a **Standard Latin Square** when the letters of the first row and first column are arranged in alphabetical order (the previous example).
- Exists an unique standard  $3 \times 3$  Latin Square, however there are four standard  $4 \times 4$  Latin Square.

| Square 1 |   |   |   | Square 2 |   |   |   | Square 3 |   |   |   | Square 4 |   |   |   |
|----------|---|---|---|----------|---|---|---|----------|---|---|---|----------|---|---|---|
| A        | B | C | D | A        | B | C | D | A        | B | C | D | A        | B | C | D |
| B        | C | D | A | B        | A | D | C | B        | A | D | C | B        | D | A | C |
| C        | D | A | B | C        | D | A | B | C        | D | B | A | C        | A | D | B |
| D        | A | B | C | D        | C | B | A | D        | C | A | B | D        | C | B | A |

# Creating a Latin Square design

- Write down any Latin square of the required size (it could be a standard Latin square).
- And randomize:
  - ▣ Randomize the order of the rows.
  - ▣ Randomize the order of the columns.
  - ▣ Randomize the allocation of treatments to the letters of the square.

# The model

- $Y_{ijt}} = \mu + R_i + C_j + T_k + \varepsilon_{ijk}$
- Where
- $Y_{ijt}$  denoting any observation for which
  - ▣  $X_1 = i, X_2 = j, X_3 = k$
  - ▣  $X_1$  and  $X_2$  are blocking factors
  - ▣  $X_3$  is the primary factor (treatment).
- $\mu$  denoting the general location parameter
- $R_i$  denoting the effect for block “i”
- $C_j$  denoting the effect for block “j”
- $T_k$  denoting the effect for treatment “k”



# Calculating the model

- Estimate for  $\mu$ :  $\bar{Y}$  = the average of all the data
- Estimate for  $R_i$ :  $\bar{Y}_i - \bar{Y}$ 
  - ▣  $\bar{Y}_i$  = average of all  $Y$  for which  $X_1 = i$
- Estimate for  $C_j$ :  $\bar{Y}_j - \bar{Y}$ 
  - ▣  $\bar{Y}_j$  = average of all  $Y$  for which  $X_2 = j$
- Estimate for  $T_k$ :  $\bar{Y}_k - \bar{Y}$ 
  - ▣  $\bar{Y}_k$  = average of all  $Y$  for which  $X_3 = k$
- Estimate for  $\varepsilon_{ijk}$ 
  - ▣  $\varepsilon_{ijk} = (y_{ijk} - \bar{y}_i - \bar{y}_j - \bar{y}_k + 2\bar{y})$

# The ANOVA table

| Source     | SS    | df (N-1)=k <sup>2</sup> -1 | MS                 | F           |
|------------|-------|----------------------------|--------------------|-------------|
| Row        | SSROW | k-1                        | MSROW = SSROW / df | MSROW / MSE |
| Column     | SSCOL | k-1                        | MSCOL = SSCOL / df | MSCOL / MSE |
| Treatments | SSTR  | k-1                        | MSTR = SSTR / df   | MSTR / MSE  |
| Error      | SSE   | (k-1)(k-2)                 | MSE = SEE / df     |             |

$$\begin{aligned}
 \underbrace{\sum_{j=1}^k \sum_{i=1}^k (y_{ijt} - \bar{y})^2}_{\text{TSS}} &= \underbrace{k \sum_{i=1}^k (\bar{y}_{i..} - \bar{y})^2}_{\text{SSROW}} + \underbrace{k \sum_{j=1}^k (\bar{y}_{.j.} - \bar{y})^2}_{\text{SSCOL}} + \underbrace{k \sum_{t=1}^k (\bar{y}_{..t} - \bar{y})^2}_{\text{SSTR}} + \\
 &\quad + \underbrace{\sum_{i=1}^k \sum_{j=1}^k \sum_{t=1}^k (y_{ijt} - \bar{y}_{i..} - \bar{y}_{.j.} - \bar{y}_{..t} + 2\bar{y})^2}_{\text{SSE}}
 \end{aligned}$$

# Example

- A courier company is interested in deciding between five brands (D,P,F,C and R) of car for its next purchase of fleet cars.
  - ▣ The brands are all comparable in purchase price.
  - ▣ The company wants to carry out a study that will enable them to compare the brands with respect to operating costs.
  - ▣ For this purpose they select five drivers (Rows).
  - ▣ In addition the study will be carried out over a five week period (Columns = weeks).

# Example

- Each week a driver is assigned to a car using randomization and a Latin Square Design.
- The average cost per mile is recorded at the end of each week and is tabulated below:

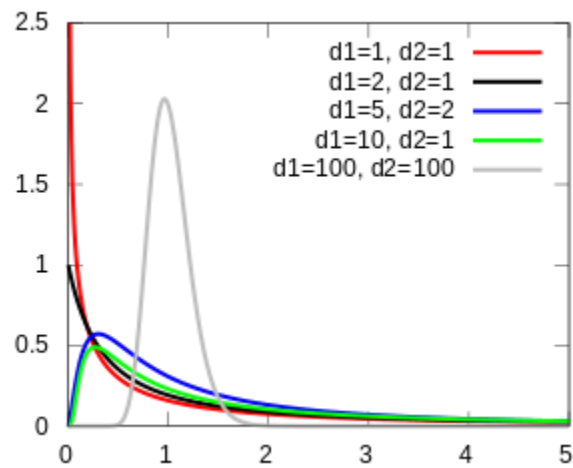
|         | Week |       |       |       |       |       |
|---------|------|-------|-------|-------|-------|-------|
|         |      | 1     | 2     | 3     | 4     | 5     |
| Drivers | 1    | 5.83  | 6.22  | 7.67  | 9.43  | 6.57  |
|         |      | D     | P     | F     | C     | R     |
|         | 2    | 4.80  | 7.56  | 10.34 | 5.82  | 9.86  |
|         |      | P     | D     | C     | R     | F     |
|         | 3    | 7.43  | 11.29 | 7.01  | 10.48 | 9.27  |
|         |      | F     | C     | R     | D     | P     |
|         | 4    | 6.60  | 9.54  | 11.11 | 10.84 | 15.05 |
|         |      | R     | F     | D     | P     | C     |
|         | 5    | 11.24 | 6.34  | 11.30 | 12.58 | 16.04 |
|         |      | C     | R     | P     | F     | D     |

# Example

## □ The Anova Table for Example

| Source        | S.S.      | d.f. | M.S.     | F ratio | F (table<br>$\alpha=0.05\%$ ) |
|---------------|-----------|------|----------|---------|-------------------------------|
| <b>Week</b>   | 51.17887  | 4    | 12.79472 | 16.06   | F(4,12)=3.25                  |
| <b>Driver</b> | 69.44663  | 4    | 17.36166 | 21.79   |                               |
| <b>Car</b>    | 70.90402  | 4    | 17.72601 | 22.24   |                               |
| <b>Error</b>  | 9.56315   | 12   | 0.79693  |         |                               |
| <b>Total</b>  | 201.09267 | 24   |          |         |                               |

# F-table

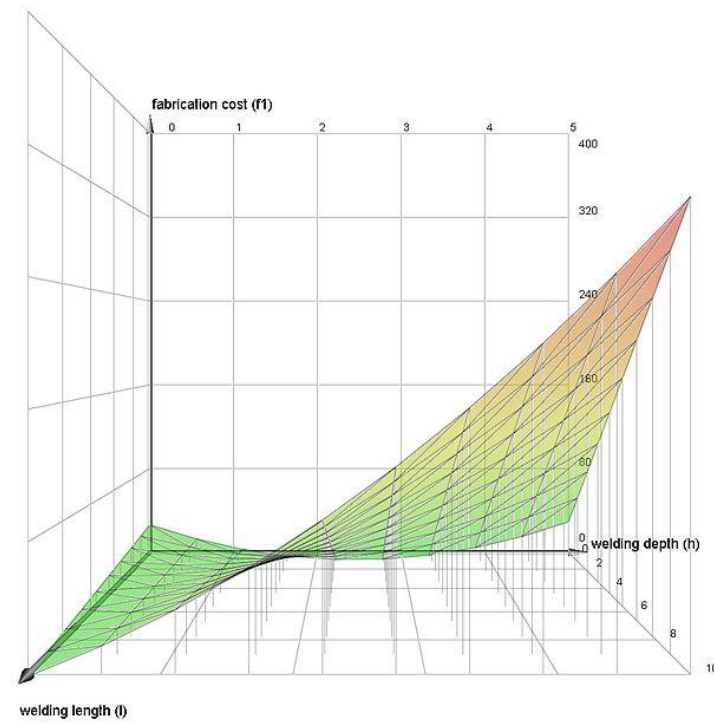
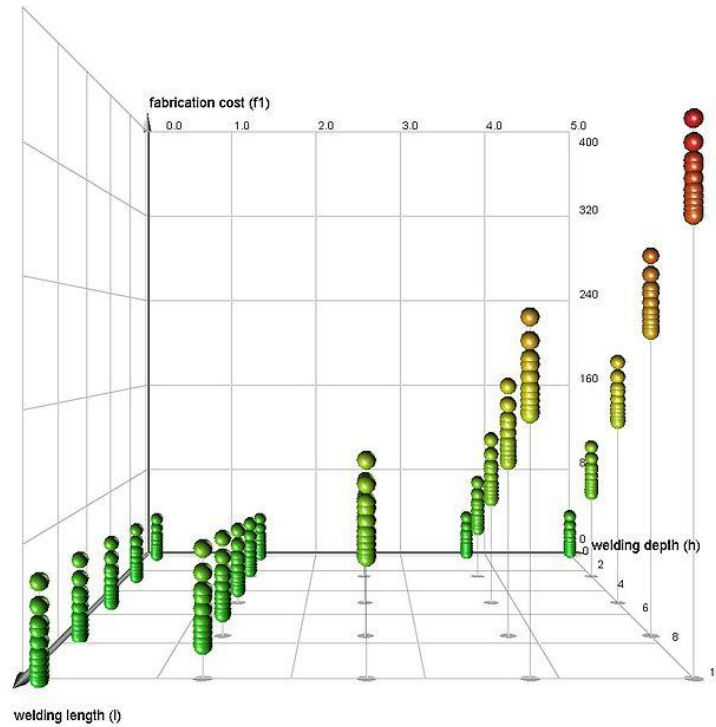


|        | Numerator df: $df_B$ |      |      |      |      |
|--------|----------------------|------|------|------|------|
| $df_W$ | 1                    | 2    | 3    | 4    | 5    |
| 5 5%   | 6.61                 | 5.79 | 5.41 | 5.19 | 5.05 |
| 1%     | 16.3                 | 13.3 | 12.1 | 11.4 | 11.0 |
| 10 5%  | 4.96                 | 4.10 | 3.71 | 3.48 | 3.33 |
| 1%     | 10.0                 | 7.56 | 6.55 | 5.99 | 5.64 |
| 12 5%  | 4.75                 | 3.89 | 3.49 | 3.26 | 3.11 |
| 1%     | 9.33                 | 6.94 | 5.95 | 5.41 | 5.06 |
| 14 5%  | 4.60                 | 3.74 | 3.34 | 3.11 | 2.96 |
| 1%     | 8.86                 | 6.51 | 5.56 | 5.04 | 4.70 |



# Response surface methods

# From full factorial design to the surface





# Steps

---

1. Set objectives
2. Select process variables
3. Select an experimental design
4. Execute the design
5. Check that the data are consistent with the experimental assumptions
6. Analyze and interpret the results
7. Use/present the results (may lead to further runs or DOE's).

# Planning and running DOE

- ❑ Check performance of gauges/measurement devices first.
- ❑ Keep the experiment as simple as possible.
- ❑ Check that all planned runs are feasible.
- ❑ Watch out for process drifts and shifts during the run.
- ❑ Avoid unplanned changes (e.g., swap operators at halfway point).
- ❑ Allow some time (and back-up material) for unexpected events.
- ❑ Obtain buy-in from all parties involved.
- ❑ Maintain effective ownership of each step in the experimental plan.
- ❑ Preserve all the raw data--do not keep only summary averages!
- ❑ Record everything that happens.
- ❑ Reset equipment to its original state after the experiment.



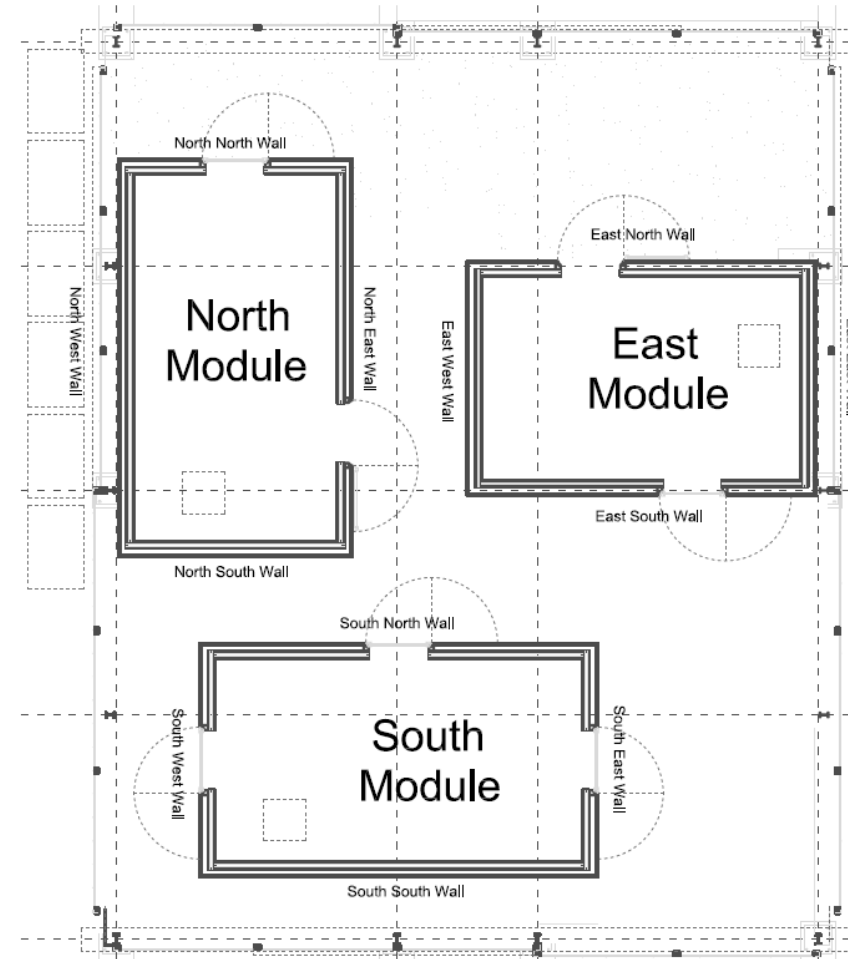
# Case study

DOE applied to construction problems

# Solar Decathlon Europe, LOW3

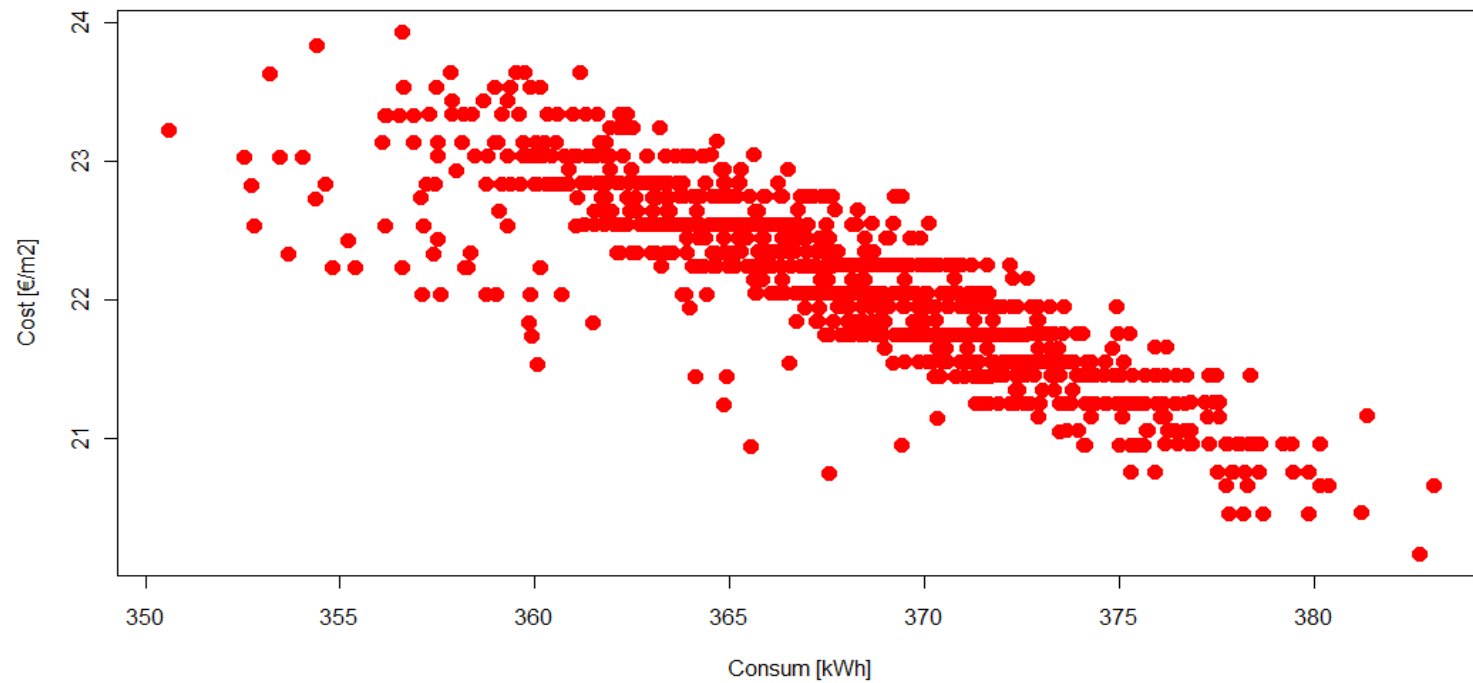


# Solar Decathlon Europe, LOW3



[illegible][illegible]

# Solar Decathlon Europe, LOW3



# Solar Decathlon Europe, LOW3

## □ Using Yates

| Module | Wall       | Effects | Mean |
|--------|------------|---------|------|
| East   | East Wall  | 4,35    | 3,34 |
|        | South Wall | 3,09    |      |
|        | West Wall  | 3,26    |      |
|        | North Wall | 2,66    |      |
| South  | East Wall  | 2,23    | 2,54 |
|        | South Wall | 2,16    |      |
|        | West Wall  | 3,44    |      |
|        | North Wall | 2,35    |      |
| North  | East Wall  | 2,74    | 3,46 |
|        | South Wall | 4,65    |      |
|        | West Wall  | 1,60    |      |
|        | North Wall | 4,86    |      |



# Solar Decathlon Europe, LOW3

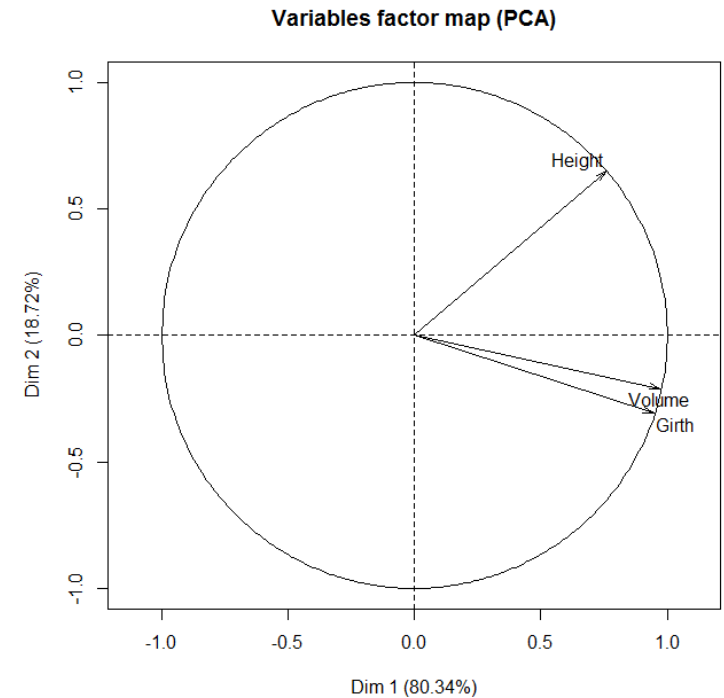
- To know more:

- ▣ Fonseca i Casas, P., Fonseca i Casas, A., Garrido-Soriano, N., & Casanovas, J. (2014). Formal simulation model to optimize building sustainability. *Advances in Engineering Software*, 69, 62–74.  
doi:10.1016/j.advengsoft.2013.12.009

# Sequoias

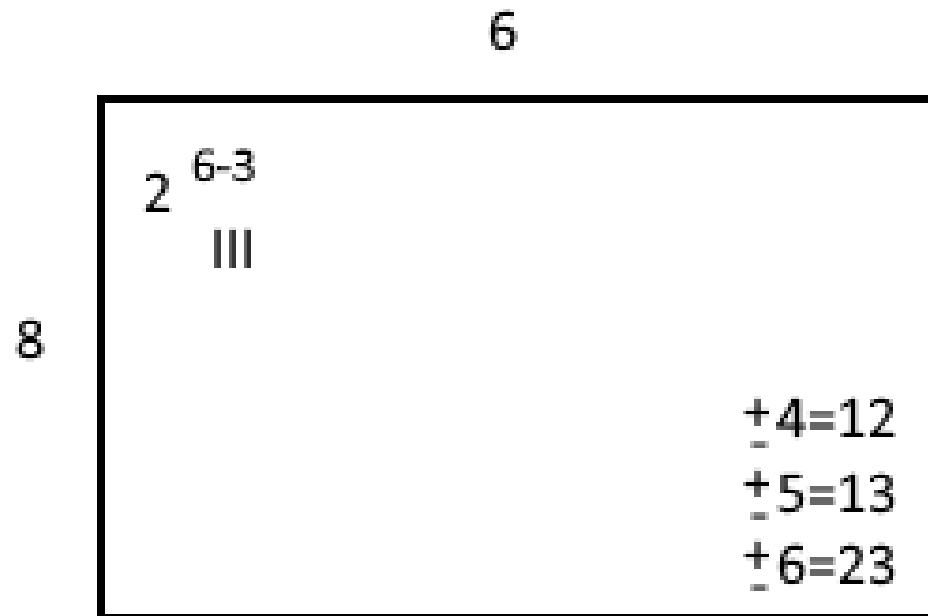
- We want to analyze the relation of the Girth, Height and the volume in large trees (sequoias). Discuss the nature of the information we have. See the table.P2 Analyzing the next diagram, what can you understand of the Volume of a tree? Describe the information you can obtain form the analysis.

|    | Girth | Height | Volume |
|----|-------|--------|--------|
| 1  | 8.3   | 70     | 10.3   |
| 2  | 8.6   | 65     | 10.3   |
| 3  | 8.8   | 63     | 10.2   |
| 4  | 10.5  | 72     | 16.4   |
| 5  | 10.7  | 81     | 18.8   |
| 6  | 10.8  | 83     | 19.7   |
| 7  | 11.0  | 66     | 15.6   |
| 8  | 11.0  | 75     | 18.2   |
| 9  | 11.1  | 80     | 22.6   |
| 10 | 11.2  | 75     | 19.9   |
| 11 | 11.3  | 79     | 24.2   |
| 12 | 11.4  | 76     | 21.0   |
| 13 | 11.4  | 76     | 21.4   |
| 14 | 11.7  | 69     | 21.3   |
| 15 | 12.0  | 75     | 19.1   |
| 16 | 12.9  | 74     | 22.2   |
| 17 | 12.9  | 85     | 33.8   |
| 18 | 13.3  | 86     | 27.4   |
| 19 | 13.7  | 71     | 25.7   |
| 20 | 13.8  | 64     | 24.9   |
| 21 | 14.0  | 78     | 34.5   |
| 22 | 14.2  | 80     | 31.7   |
| 23 | 14.5  | 74     | 36.3   |
| 24 | 16.0  | 72     | 38.3   |
| 25 | 16.3  | 77     | 42.6   |
| 26 | 17.3  | 81     | 55.4   |
| 27 | 17.5  | 82     | 55.7   |
| 28 | 17.9  | 80     | 58.3   |
| 29 | 18.0  | 80     | 51.5   |
| 30 | 18.0  | 80     | 51.0   |
| 31 | 20.6  | 87     | 77.0   |



# Sequoias

- We want to conduct an experiment to understand if we can modify the climatic condition (mainly gases (mainly  $\text{CO}_2$  and  $\text{H}_2\text{O}$ ) and these factors (Volume, Temperature, Humidity, etc.). The main idea is to plant those trees in different environments and analyze the evolution of those trees for 8 different environments.
- What will be the experimental design to be used? Justify your answers.
- How are you going to deal with randomness?



# yogurt

- Imagine you're a food engineer and you want to optimize the process of making a new type of yogurt. You have five factors that you think can influence the final quality of the product:
  - A: Bacterial Culture Type (2 Levels: A, A)
  - B: Fermentation time (2 levels: B, b)
  - C: Fermentation temperature (2 levels: C, c)
  - D: Amount of sugar (2 levels: D, d)
  - E: Fruit Type (2 levels: E, e)
- Due to time and resource constraints, we can only run 8 scenarios.

# yogurt

| A | B | C | D | E | Answer |
|---|---|---|---|---|--------|
| - | - | - | - | - | 10     |
| + | - | - | - | - | 12     |
| - | + | - | - | - | 14     |
| + | + | - | - | - | 16     |
| - | - | + | - | - | 18     |
| + | - | + | - | - | 20     |
| - | + | + | - | - | 22     |
| + | + | + | - | - | 24     |
| - | - | - | + | - | 26     |
| + | - | - | + | - | 28     |
| - | + | - | + | - | 30     |
| + | + | - | + | - | 32     |
| - | - | + | + | - | 34     |
| + | - | + | + | - | 36     |
| - | + | + | + | - | 38     |
| + | + | + | + | - | 40     |

|   |   |   |   |   |    |
|---|---|---|---|---|----|
| - | - | - | - | + | 42 |
| + | - | - | - | + | 44 |
| - | + | - | - | + | 46 |
| + | + | - | - | + | 48 |
| - | - | + | - | + | 50 |
| + | - | + | - | + | 52 |
| - | + | + | - | + | 54 |
| + | + | + | - | + | 56 |
| - | - | - | + | + | 58 |
| + | - | - | + | + | 60 |
| - | + | - | + | + | 62 |
| + | + | - | + | + | 64 |
| - | - | + | + | + | 66 |
| + | - | + | + | + | 68 |
| - | + | + | + | + | 70 |
| + | + | + | + | + | 72 |

# yogurt

| Número de factores, k | Fracción  | Número de combinaciones | Generadores de diseño |
|-----------------------|-----------|-------------------------|-----------------------|
| 3                     | $2^{3-1}$ | 4                       | C=AB                  |
| 4                     | $2^{4-1}$ | 8                       | D=ABC                 |
| 5                     | $2^{5-1}$ | 16                      | E=ABCD                |
|                       | $2^{5-2}$ | 8                       | D=AB E=AC             |
|                       | $2^{6-1}$ | 32                      | F=ABCDE               |
| 6                     | $2^{6-2}$ | 16                      | E=AB                  |
|                       | $2^{6-3}$ | 8                       | D=AB<br>E=AC          |

1. D = AB

2. E = AC

| A  | B  | C  | D  | E  | Answer |
|----|----|----|----|----|--------|
| -1 | -1 | -1 | 1  | 1  | 58     |
| 1  | -1 | -1 | -1 | -1 | 12     |
| -1 | 1  | -1 | -1 | 1  | 46     |
| 1  | 1  | -1 | 1  | -1 | 32     |
| -1 | -1 | 1  | 1  | -1 | 34     |
| 1  | -1 | 1  | -1 | 1  | 52     |
| -1 | 1  | 1  | -1 | -1 | 22     |
| 1  | 1  | 1  | 1  | 1  | 72     |

Yates

| 1   | 2   | 3   | Contrasts |
|-----|-----|-----|-----------|
| 70  | 148 | 328 | 41 Mean   |
| 78  | 180 | 8   | 2 A       |
| 86  | -60 | 16  | 4 B       |
| 94  | 68  | 64  | 16 D      |
| -46 | 8   | 32  | 8 C       |
| -14 | 8   | 128 | 32 E      |
| 18  | 32  | 0   | 0         |
| 50  | 32  | 0   | 0         |

EffA= 2  
 EffB= 4  
 EffC= 8  
 EffD= 16  
 EffE= 32

# Fast experiment

- We have 3 factors, as defined below, but we want to execute only 4 scenarios. Calculate the design, and the main effects for A, B and C.

| A | B | C |    |
|---|---|---|----|
| - | - | - | 10 |
| + | - | - | 20 |
| - | + | - | 30 |
| + | + | - | 40 |
| - | - | + | 50 |
| + | - | + | 60 |
| - | + | + | 70 |
| + | + | + | 80 |





# Example: $2^3$ Factorial in 2 Blocks (each of size 4)

- Design: 3 factors (A, B, C), 8 runs total.
  - ▣ We want 2 blocks of 4 runs.
- **Confound ABC with blocks**
  - ▣ Block 1: runs where  $ABC = +$
  - ▣ Block 2: runs where  $ABC = -$

## Example: $2^3$ Factorial in 2 Blocks (each of size 4)

- **Result:** The 3-factor interaction ABC is **confounded with blocks**. Main effects (A, B, C) and 2-factor interactions (AB, AC, BC) are **free** of block effects.

| Block | A | B | C | ABC | Response |
|-------|---|---|---|-----|----------|
| 1     | + | + | + | +   | y1       |
| 1     | + | - | - | +   | y2       |
| 1     | - | + | - | +   | y3       |
| 1     | - | - | + | +   | y4       |
| 2     | - | - | - | -   | y5       |
| 2     | - | + | + | -   | y6       |
| 2     | + | - | + | -   | y7       |
| 2     | + | + | - | -   | y8       |